Can Agentic AI Settle a 20-Year-Old Description Logic Problem?

On the Decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$, a New Decidable Fragment, and a Certified Lean Formalization Reducing It to One Keystone

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Abstract

Whether concept satisfiability in $\mathcal{ALCI}_{\mathrm{RCC5}}$—the description logic $\mathcal{ALCI}$ with the RCC5 base relations as roles under *abstract composition-table* semantics—is decidable has been open since Wessel's 2002/2003 work. This paper reports a sustained, adversarially-reviewed, partly machine-checked attack on the problem, conducted by AI assistants under human direction. We do not settle the question. What we establish is: (i) a *conditional decidability theorem*—if the "live width" of models is bounded by a computable function of the concept (property F6), then satisfiability is decidable—whose soundness direction is *machine-checked in Lean 4*; (ii) that the entire remaining difficulty compresses into that single property F6, which we show is the *same fact* that blocks a proof of undecidability; (iii) an RCC5 normal form (forward direction machine-certified) and an unconditionally decidable fragment (the $\forall\mathsf{PO}$-free concepts); (iv) a *working, empirically-validated prototype reasoner*—a cover-tree tableau that reproduces, 23 years on, the GIS concept taxonomy of the original report, cross-validated on 911 concepts with zero mismatches and never once contradicted by an independent oracle or a hand analysis; and (v) an honest map of the routes tried, the dead ends, and the missing hard pieces. We also report, candidly, on the methodology—adversarial "cold review," machine verification, and a human-in-the-loop workflow—and on what frontier AI could and could not do.

Keywords: description logics; $\mathcal{ALCI}$; RCC5; region connection calculus; qualitative spatial reasoning; decidability; concept satisfiability; patchwork property; interactive theorem proving (Lean); AI-assisted mathematics.

ACM CCS: Theory of computation → Description logics; Automated reasoning; Logic and verification. Computing methodologies → Spatial and physical reasoning.

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Contents

1 Introduction

1.1 AI disclaimer and authorship

The mathematical content of this paper was produced by AI assistants (Anthropic's Claude models and OpenAI's GPT-5.5), prompted, steered, and reviewed by the human author (Michael Wessel), who posed the problem, supplied the central intuitions and directions, orchestrated the review–repair cycles, and takes responsibility for the framing. **The results are unverified by human domain experts and should be read as AI-generated conjectures supported by machine checking and computational evidence—not as established theorems**, with the explicit exceptions of the parts stated to be kernel-checked in Lean 4 (the soundness pipeline, the abstraction's faithfulness, and the RCC5 normal form), which are machine-verified but still human-unreviewed. We are deliberate about this labelling throughout: the standing one-line status is *strongly supported, not certified*. We regard the AI systems as having made contributions of a kind that, in a human collaborator, would warrant co-authorship, and we discuss this in §9.

1.2 Why the problem is hard, in one page

RCC5 relates two regions by exactly one of five relations: equal (EQ), proper part (PP), proper superpart (PPI), partial overlap (PO), or discreteness/disjointness (DR). Under *abstract composition-table* semantics a model is not a set of actual regions but a set with a total labelling of ordered pairs by these relations, required only to be converse-closed and *composition-closed*: if $x\,R\,y$ and $y\,S\,z$ then $x\,T\,z$ for some T the composition table allows.

The five relations split into two axes that behave completely differently.

- **The vertical axis** ($\mathsf{PP}, \mathsf{PPI}$: nesting) is *tame*. Composition can *force* vertical structure—the only four "determining" compositions (those returning a single relation) all run through PP or PPI—but an unbounded vertical chain is captured by a finite *generator*: a self-loop that *unfolds* into infinitely many distinct points (never an actual PP-cycle, which strong EQ forbids), just as regular structures capture infinite paths in modal logic. Infinite depth is not the problem.

- **The horizontal axis** (DR, PO: side-by-side) is *wild*. No *single* composition step pins a horizontal relation to one value (PO is never a singleton table entry)—though as constraints accumulate, their *intersection* sometimes can. What composition still cannot do is force two occurrences *together*; it can only force them *apart* (a shared horizontal anchor excludes EQ). So horizontal "crowds" of pairwise-distinct occurrences are the danger.

A finite decision procedure must record, at each interface of a model, the genuine branching it cannot reconstruct—the *live width*. Many forced edges are *shadows* (implied by composition from edges already present) and cost nothing; live width counts only the rest. The whole decidability question reduces to one property:

> **(F6)** Is the live width of models bounded by a computable function of the input concept?

If yes, the certificate space is finite and satisfiability is decidable. And here is the knife's edge: F6 is *the same fact* that blocks proving the problem *undecidable*. To force undecidability one would encode a rigid coordinate grid, which requires forcing a horizontal relation to a single value—exactly what the composition table cannot do (the "coincidence obstruction" Wessel identified in 2002/2003). Turn that looseness into a bound and one gets decidability; circumvent it and force a grid and one gets undecidability. The two open directions are one question seen from two sides, which is why neither has been settled in over twenty years.

1.3 Scope, contributions, and structure

This is both a technical report and a discussion piece. Its contributions are:

1. A *conditional decidability theorem* (F6 $\Rightarrow$ decidable) whose soundness half is machine-checked in Lean 4 (§7.5), and the reduction of the whole problem to F6.

2. The identification of F6 as the shared keystone of the decidability and undecidability directions (§1.2, §10).

3. An RCC5 *normal form* (forward direction machine-certified) and an unconditionally decidable $\forall\mathsf{PO}$-*free* fragment (§7.2, §7.6).

4. An honest, technique-organized account of every route tried, the counterexamples that killed the wrong ideas, and the lessons (§7), so that others need not repeat them.

5. A candid methodology report on adversarial AI-assisted mathematics (§9).

The paper is organized as follows. §2 states, in largely non-technical terms, where the problem stands after twenty-five years. §4 gives the history ($\mathcal{ALCI}_{\mathrm{RCC}}$ from Cohn 1993 through Wessel 2002/2003 and Lutz–Wolter 2006) and carefully distinguishes our object from the neighbouring logics that *are* known undecidable. §5 views the problem as first-order satisfiability and asks whether it falls inside a decidable fragment of first-order logic—it does not, and the deepest reason (totality) previews the central obstruction. §6 explains why the standard undecidability reductions fail. §7—the core of the paper—surveys the approaches *by technique*: naive tableaux and blocking, finite-model and reduction attempts, the split-forest idea, the automata route, the Lean line culminating in F6, and the regular-cover pivot. §8 describes the working reasoners and a GIS example. §9 reports on the AI methodology. §10 concludes with what was learned, what would move the needle, an honest estimate of the chances of closing F6, and the project's actual wins.

2 Current status: where we stand after twenty-five years

Decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept satisfiability remains open (the closely related $\mathcal{ALCI}_{\mathrm{RCC8}}$ likewise, but it is not our focus here). A current-literature check (July 2026) found no published resolution; the closest neighbours are decided in the other direction and are discussed in §4. What the present work adds is not an answer but a sharp reduction and a machine-checked surround.

The positive result—conditional decidability. *If* the live width of models is bounded by a computable function of the concept (property F6), *then* concept satisfiability is decidable; and the **soundness half of that reduction is machine-checked in Lean 4** (a valid finite certificate provably unfolds to a genuine RCC5 model of the concept, and the finite type abstraction is faithful). What is open is the antecedent F6 itself—equivalently the completeness direction, that every satisfiable concept has a *bounded* finite certificate.

What is certified (machine-checked, Lean 4, zero `sorry`):

- the *soundness* pipeline: a valid finite certificate unfolds to a genuine RCC5 model of the concept;
- the *faithfulness* of the finite (Hintikka) abstraction: satisfiability and abstraction-realizability coincide;
- the *RCC5 normal form* (forward): every strong-EQ RCC5 network is an ordered-disjoint structure—a strict partial order for PP with a downward-closed disjointness for DR, PO residual.

What is strongly supported but not certified: F6, the single open keystone—now sharpened to a precise combinatorial dichotomy (does some concept force unbounded "identity-selector minors"?), with the auxiliary uniformization property W2′ shown to *fold into* it.

Unconditional by-products (independent of F6):

- the $\forall\mathsf{PO}$-free fragment is *decidable*—concepts with no $\forall\mathsf{PO}$ subformula, still retaining $\exists\mathsf{PO}$, $\forall\mathsf{DR}$, $\exists\mathsf{DR}$, and all vertical modalities, so a genuinely expressive spatial fragment;

- the RCC5 normal form above (forward direction certified);
- a working, empirically validated *prototype reasoner* (the cover-tree tableau; no completeness theorem is claimed): cross-validated on 911 concepts against an independent oracle with zero mismatches, reproducing 23 years later the full GIS subsumption taxonomy of Wessel's original report, and never once contradicted—no counterexample to its verdicts was found anywhere in the campaign;
- the standard undecidability routes shown blocked, with the coincidence obstruction identified as the shared fact.

The honest bottom line. After roughly thirty repair rounds and sixteen adversarial reviews the project has reached a genuine *local optimum*: the local algebra is exhausted (and in one case certified), the soundness side is machine-checked, and the whole remaining difficulty is one well-posed lemma. We offer the work not as a resolution but as a two-decade open problem *reduced, honestly and checkably, to its irreducible core.*

3 Preliminaries

We fix the object logic precisely; readers familiar with $\mathcal{ALCI}_{\mathrm{RCC}}$ may skim to Definition 5.

Syntax. $\mathcal{ALCI}_{\mathrm{RCC5}}$ concepts are built from atomic concepts A and the five RCC5 relations, used as roles, by

$$C ::= A \mid \neg A \mid \top \mid \bot \mid C \sqcap C \mid C \sqcup C \mid \exists R.C \mid \forall R.C, \qquad R \in \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}.$$

We work in *negation normal form* (NNF: negation only on atoms) and *absorb inverse roles* through the RCC5 converse—$\exists \mathsf{PP}^{-}.C = \exists \mathsf{PPI}.C$, $\exists \mathsf{PO}^{-}.C = \exists \mathsf{PO}.C$, and so on—so no separate inverse constructor is needed. Let $\mathsf{cl}(C_0)$ be the Fischer–Ladner closure of the input concept and $m = |\mathsf{cl}(C_0)|$.

Semantics: abstract composition tables, strong EQ. An interpretation is a pair $\mathcal{I} = (\Delta, \rho)$ with $\Delta \neq \emptyset$ and a *total* labelling $\rho : \Delta \times \Delta \to \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$ satisfying: (i) *strong* EQ: $\rho(x, y) = \mathsf{EQ}$ iff $x = y$ (so EQ is identity, and between distinct elements only $\{\mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$ occur); (ii) *converse-closed*: $\rho(y, x) = \mathrm{conv}(\rho(x, y))$, where $\mathsf{PP}^{-1} = \mathsf{PPI}$, $\mathsf{PO}^{-1} = \mathsf{PO}$, $\mathsf{DR}^{-1} = \mathsf{DR}$; and (iii) *composition-closed*: $\rho(x, z) \in \mathsf{comp}(\rho(x, y), \rho(y, z))$ for all x, y, z, over the RCC5 composition table (Table 2). Concept extensions are standard, reading a role R as the pair set $\{(x, y) : \rho(x, y) = R\}$; the concept C_0 is *satisfiable* if $x \in C_0^{\mathcal{I}}$ for some $\mathcal{I}$ and some $x \in \Delta$. Thus a model is a complete graph carrying a composition-closed RCC5 edge-labelling; *models may be infinite*. This is precisely the *abstract* (composition-table) semantics—not a semantics of actual regions in a topological space. We fix the *strong* EQ semantics throughout (the standard convention for this problem; strong versus weak EQ is discussed below). The decision problem is: given C_0, is it satisfiable?

The sister logic $\mathcal{ALCI}_{\mathrm{RCC8}}$. $\mathcal{ALCI}_{\mathrm{RCC8}}$ is defined identically, over the eight RCC8 relations $\{\mathsf{DC}, \mathsf{EC}, \mathsf{PO}, \mathsf{EQ}, \mathsf{TPP}, \mathsf{NTPP}, \mathsf{TPPI}, \mathsf{NTPPI}\}$, which *refine* RCC5 by splitting DR into DC (disconnected) and EC (externally connected, i.e. touching at the boundary), and $\mathsf{PP}, \mathsf{PPI}$ into tangential and non-tangential parts ($\mathsf{TPP}, \mathsf{NTPP}$ and $\mathsf{TPPI}, \mathsf{NTPPI}$); PO and EQ are unchanged. Its abstract composition-table semantics is the exact analogue over the RCC8 composition table, and RCC8 likewise has the patchwork property. $\mathcal{ALCI}_{\text{RCC5}}$ is our focus; we invoke $\mathcal{ALCI}_{\text{RCC8}}$ only where it matters—its ExpTime-hardness (§4) and the domino attempts (§6), which turn on the extra $\mathsf{TPP}/\mathsf{NTPP}$ distinction.

The RCC granularity hierarchy. $\mathcal{ALCI}_{\text{RCC5}}$ and $\mathcal{ALCI}_{\text{RCC8}}$ sit atop a family of five logics $\mathcal{ALCI}_{\text{RCCk}}$, $k \in \{1,2,3,5,8\}$, whose relation sets *coarsen* as k decreases [8]: $\mathcal{ALCI}_{\text{RCC1}}$ has the single relation SR ("spatially related"); $\mathcal{ALCI}_{\text{RCC2}}$ has $\{\mathsf{DR}, \mathsf{O}\}$ (with O = "overlapping"); and $\mathcal{ALCI}_{\text{RCC3}}$ has $\{\mathsf{DR}, \mathsf{ONE}, \mathsf{EQ}\}$ (with ONE = "overlapping but not equal"). Each coarser set is definable in the finer one—$\mathsf{ONE} = \{\mathsf{PP}, \mathsf{PPI}, \mathsf{PO}\}$, $\mathsf{O} = \{\mathsf{ONE}, \mathsf{EQ}\}$, $\mathsf{SR} = \{\mathsf{DR}, \mathsf{O}\}$—so the family forms a granularity lattice. The three coarsest logics are *decidable* (Wessel [8]); $\mathcal{ALCI}_{\text{RCC5}}$ and $\mathcal{ALCI}_{\text{RCC8}}$ are the open cases this paper concerns. The full landscape is Table 1 (§4).

Strong versus weak EQ. Two semantics for EQ appear in the literature [8]: the *strong* one, $\mathsf{EQ}^{\mathcal{I}} = \mathrm{Id}(\Delta)$ (EQ is exactly identity), and the *weak* one, $\mathrm{Id}(\Delta) \subseteq \mathsf{EQ}^{\mathcal{I}}$ (EQ contains identity but may relate distinct, mutually indistinguishable elements). We fix the *strong* semantics throughout—the standard choice for this problem, and the more expressive one: Wessel observed that under strong EQ, from $\mathcal{ALCI}_{\text{RCC3}}$ upward the *difference role* $R_D =_{\text{def}} N_R \setminus \{\mathsf{EQ}\}$ becomes definable, so *nominals* are encodable ($i \sqcap \forall R_D.\neg i$ forces i to name a single element)—a reminder of how narrow the margin to undecidability is (adding a $\downarrow$-binder would cross it, cf. §4.3). The choice costs no generality for decidability: Wessel showed decidability under the weak semantics would carry over to the strong one, and conversely a weak-EQ model collapses to a strong one by internalizing EQ as identity in the TBox, so the two satisfiability problems are inter-reducible.

The two axes, made precise. Call $\mathsf{PP}, \mathsf{PPI}$ *vertical* (nesting) and DR, PO *horizontal* (side-by-side). The composition table makes the axes behave oppositely, and this asymmetry is the root of the whole difficulty.

Proposition 1 (Determination is vertical)**.** *Of the sixteen compositions of proper (non-EQ) relations, exactly four are* determining *(return a singleton):*

$$\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}, \;\; \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}, \;\; \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}, \;\; \mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}.$$

Every one has a vertical (PP or PPI) argument, and PO is never the value of any determining composition. (Exhaustive over Table 2; machine-verified.)

So no *single* composition step forces a horizontal value—the four determining steps are all vertical. This must not be overread:

Remark 2 (Single-step versus joint forcing)**.** Proposition 1 concerns *one* length-two path; it does *not* say horizontal edges are always free. The *intersection* of several composition constraints can pin a horizontal edge to a single value: $\mathsf{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PPI}, \mathsf{PO}\}$ intersect in $\{\mathsf{PO}\}$, so an edge forced through two different intermediates can be *compelled* to be PO (machine-checked, `wp85`); indeed every proper relation—$\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}$—is reachable as such a joint singleton, so PO is not special among them. Consequently "shadow" and "live" (Definition 5) must be read against the *joint* closure of all constraints, never single-path forcing. What *composition* genuinely never forces is a merge: no intersection of table cells ever narrows to $\{\mathsf{EQ}\}$ (`wp85`), so composition can only force occurrences *apart* (a shared horizontal anchor excludes EQ), never *together*. Concept-level isolation *can* still force a merge—but only self-defeatingly: the markers that would force $z = z'$ also collapse the construction to unsatisfiability (§6.2). So the coincidence obstruction is really a *dichotomy*—a would-be grid merge is either *unforced* (the branches never join) or *fatal* (it collapses to UNSAT), never a clean forced grid—not a blanket "EQ cannot be forced." That one-sidedness, forcing apart but not (usefully) together, is the real horizontal asymmetry.

There is one further asymmetry, and it is the one the split-forest presentation (§7.3) turns to its advantage: disjointness is inherited *downward* but not upward.

Proposition 3 (Half-determinism of disjointness)**.** *If $x \,\mathsf{DR}\, y$ then every part of x is disjoint from every part of y:* $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ *forces $x' \,\mathsf{DR}\, y$ for every $x' \,\mathsf{PP}\, x$, and* $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ *pushes this down the y-side as well.* Upward *the inference fails: a* superpart *of a disjoint region may be disjoint from, overlap, or contain the other region—*$\mathsf{comp}(\mathsf{PPI}, \mathsf{DR}) = \{\mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$ *is not a singleton, so nothing is forced. Disjointness thus propagates deterministically to descendants in the* PP*-order, and to nothing else—the* downward-closed disjointness *that the RCC5 normal form (Theorem 13, forward direction certified) makes exact.*

This "half-determinism" (deterministic downward, free upward) is what lets a finite presentation dispatch arbitrarily wide DR crowds for free (§7.3): a single high DR edge shadows every disjointness beneath it, leaving only PO and the *upward* direction as genuine branching. Vertical recurrence, moreover, always folds:

Lemma 4 (No horizontal recurrence)**.** *Form the composition-path subset automaton whose states are the reachable relation sets, with $\mathrm{Rel}(\varepsilon) = \{\mathsf{EQ}\}$ and $\mathrm{Rel}(w\,a) = \bigcup_{r \in \mathrm{Rel}(w)} \mathsf{comp}(r, a)$. Restrict to states not containing* EQ*, and to non-*EQ *labels (an* EQ *label is a self-loop on every state, since* EQ *is the composition identity). Then no cycle uses a* horizontal *(*DR/PO*) label: every cycle is built from vertical (*PP/PPI*) steps. Hence any support recurrence that never folds by an equality is* eventually vertical*, and is absorbed by vertical looping rather than by a horizontal crowd. (Exhaustive finite computation; machine-verified,* `wp38`*.)*

Proposition 1 and Lemma 4 explain why infinite *depth* is never the obstacle: forced structure is vertical, and vertical chains loop. Everything hard is horizontal—and, crucially, while *individual* horizontal edges can be pinned (by the joint closure of several constraints, Remark 2), no concept is known to *force an unboundedly wide rigid* horizontal pattern. Whether one can is exactly F6. A decision procedure must record only the horizontal branching it cannot reconstruct:

Definition 5 (Shadows, live width, and F6)**.** In a model, or in a finite presentation of one, an edge is a *shadow* if its RCC5 value is forced by the *joint* composition-closure of the edges already present—which may be a single triangle ($x \,\mathsf{PP}\, y$, $y \,\mathsf{DR}\, z$ force $x \,\mathsf{DR}\, z$ by $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$) *or* the intersection of several (Remark 2); a shadow costs a decision procedure nothing. The *live width* at an interface is the number of non-shadow edges there—the genuine branching that must be recorded. Property **F6** states: for every satisfiable C_0, the live width of some presentation of a model of C_0 is bounded by a computable function of C_0.

4 History of $\mathcal{ALCI}_{\mathrm{RCC}}$ and why the known results do not settle it

4.1 Cohn's 1993 proposal

The idea of combining description logics with qualitative spatial reasoning via modal accessibility relations was first proposed by Cohn [1], who considered treating the RCC8 relations (or subsets thereof) as modalities in a multi-modal logic. The framework allowed spatial constraints to be expressed as modal formulas—e.g., $\langle\mathsf{PP}\rangle C$ for "some proper part satisfies C"—but Cohn left all decidability and complexity questions open.

Logic	Base relations N_R	Concept satisfiability
$\mathcal{ALCI}$ (no RCC)	arbitrary roles	PSPACE-complete (bare); EXPTIME w/ TBoxes
$\mathcal{ALCI}_{\text{RCC1}}$	$\{\mathsf{SR}\}$	decidable; NP-complete ($\equiv$ S5)
$\mathcal{ALCI}_{\text{RCC2}}$	$\{\mathsf{DR}, \mathsf{O}\}$	decidable; NP-hard, in PSPACE
$\mathcal{ALCI}_{\text{RCC3}}$	$\{\mathsf{DR}, \mathsf{ONE}, \mathsf{EQ}\}$	decidable; NP-hard
$\mathcal{ALCI}_{\text{RCC5}}$, $\forall$PO-free	RCC5 without $\forall$PO	**decidable** (this paper, Thm 10)
$\mathcal{ALCI}_{\text{RCC5}}$ (full)	$\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$	**open**; PSPACE-hard
$\mathcal{ALCI}_{\text{RCC8}}$	RCC8 (eight relations)	**open**; EXPTIME-hard

Table 1: The complexity landscape of the $\mathcal{ALCI}_{\text{RCC}}$ family (concept satisfiability, strong EQ). RCCk relation sets coarsen as k decreases; $\mathcal{ALCI}_{\text{RCC1}}$–$\mathcal{ALCI}_{\text{RCC3}}$ were shown decidable by Wessel [8], while $\mathcal{ALCI}_{\text{RCC5}}$ and $\mathcal{ALCI}_{\text{RCC8}}$ are the standing open cases.

4.2 Wessel's $\mathcal{ALCI}_{\text{RCC}}$ family (2002/2003)

Wessel [7, 8, 5] gave the first rigorous treatment of the combination, introducing the $\mathcal{ALCI}_{\text{RCC}}$ *family*: description logics $\mathcal{ALCI}_{\text{RCCk}}$ ($k = 1, 2, 3, 5, 8$) extending $\mathcal{ALCI}$ with role boxes derived from the RCC composition tables. In an $\mathcal{ALCI}_{\text{RCCk}}$ interpretation, every pair of domain elements is assigned exactly one RCCk base relation, and the composition table is enforced globally (*strong* EQ *semantics*: EQ is identity). Models are complete graphs with RCCk edge-colorings.

Wessel established the decidability and complexity landscape summarized in Table 1. The three coarsest logics are *decidable*: $\mathcal{ALCI}_{\text{RCC1}}$ is essentially the modal logic S5 (hence NP-complete); $\mathcal{ALCI}_{\text{RCC2}}$ and $\mathcal{ALCI}_{\text{RCC3}}$ are decidable because their composition tables are deterministic or near-deterministic, admitting reduction to standard DL techniques (for $\mathcal{ALCI}_{\text{RCC3}}$ under strong EQ, via the definable difference role). The two finest logics—$\mathcal{ALCI}_{\text{RCC5}}$ (PSPACE-hard) and $\mathcal{ALCI}_{\text{RCC8}}$ (EXPTIME-hard)—were left as the central open problem.

In a series of companion reports, Wessel also mapped out the decidability landscape for description logics with general composition-based role inclusion axioms:

- $\mathcal{ALC}_{\mathcal{RA}}$ (arbitrary role axioms) is **undecidable**, via reduction from the Post Correspondence Problem [6].
- $\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$ (non-disjoint-roles variant) is also **undecidable**, via reduction from non-empty CFG intersection [3].
- $\mathcal{ALC}_{\mathcal{RA}_{SG}}$ (admissible role boxes: deterministic, functional, complete, and associative) is **decidable** (EXPTIME-complete, via reduction to $\mathcal{ALC}$ with general TBoxes) [4]; adding unqualified number restrictions ($\mathcal{ALCN}_{\mathcal{RA}_{SG}}$) makes it undecidable again.

$\mathcal{ALCI}_{\text{RCC5}}$ sits precisely in the gap: its role box is non-deterministic (unlike $\mathcal{ALC}_{\mathcal{RA}_{SG}}$) but has the patchwork property (unlike $\mathcal{ALC}_{\mathcal{RA}}/\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$).

Remark 6 (Key difficulties)**.** $\mathcal{ALCI}_{\text{RCC5}}$ has none of the properties that standard DL decidability proofs rely on: (i) no tree model property (models are complete graphs), (ii) no finite model property (some satisfiable concepts require infinite models), (iii) a non-deterministic role box (multi-valued composition entries).

For reference, the full RCC5 composition table is given in Table 2. Reading: row $= S(b, c)$, column $= R(a, b)$, entry $=$ possible relations for (a, c).

comp	DR(a,b)	PO(a,b)	EQ(a,b)	PPI(a,b)	PP(a,b)
DR(b,c)	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
PO(b,c)	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
EQ(b,c)	DR	PO	EQ	PPI	PP
PP(b,c)	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
PPI(b,c)	DR	DR, PO, PPI	PPI	PPI	$*$

Table 2: The RCC5 composition table ($*$ = all five relations).

Key composition facts exploited in the cover-tree theory (§7.3, §8): $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ (DR rigidity); $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ (transitivity); and $\mathsf{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ (PP can be *forced* by composition and universal filtering; see the forced-composition examples in §7.1).

4.3 Wessel's grid constructions and the coincidence obstruction

In his investigation of $\mathcal{ALCI}_{\text{RCC8}}$ [8], Wessel made two important discoveries:

1. Even-odd chains (Section 4.4.2). The TPP/NTPP distinction in RCC8 allows the construction of *functional successor chains*: each node alternates between *even* and *odd*, and a node can distinguish its direct TPPI-successor from all indirect ones by parity. Wessel used this to encode a binary n-bit counter, proving $\mathcal{ALCI}_{\text{RCC8}}$ is ExpTime-hard.

2. The grid coincidence obstruction (Section 4.4.3, Figure 11). Wessel constructed explicit RCC8 networks isomorphic to $n \times n$ grids, using TPP for horizontal and vertical successors. However, he identified a fundamental obstacle to encoding domino tilings: the *coincidence condition* $R_X \circ R_Y = R_Y \circ R_X$ (east-of-north equals north-of-east) cannot be enforced by concept terms. The composition table allows $\{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$ at the coincidence point—too many options. Wessel concluded:

> "Even though these infinite models do exist, we have found no way to *enforce* the depicted model(s) by means of a concept term."

Wessel further observed that adding a hybrid logic binding operator $\downarrow$ (nominals) would immediately yield undecidability, confirming that the coincidence condition is the *precise boundary*. His explicit grid frame is reproduced here as Figure 1, and the merge obstruction that blocks enforcing it as Figure 2.

Wessel thus independently developed, three to four years before Lutz–Wolter, the RCC8 description-logic analogues of these ingredients—the even-odd chain, an explicit two-dimensional TPP/NTPP grid, and the first identification of the coincidence gap as the decisive obstacle. Lutz–Wolter later obtained undecidability by a *different* route—a linearized tiling enumeration (their §5, following Marx–Reynolds) that sidesteps the two-dimensional coincidence rather than enforcing it—so this is shared lineage and prior identification, not a dependence of their proof on Wessel's constructions.

4.4 Lutz & Wolter (2006)

Lutz and Wolter [9] studied *modal logics of topological relations*, denoted L_{RCC8}, L_{RCC5}, etc. These have the same *syntax* as $\mathcal{ALCI}_{\text{RCC}}$ but are interpreted over *topological models*: the domain elements

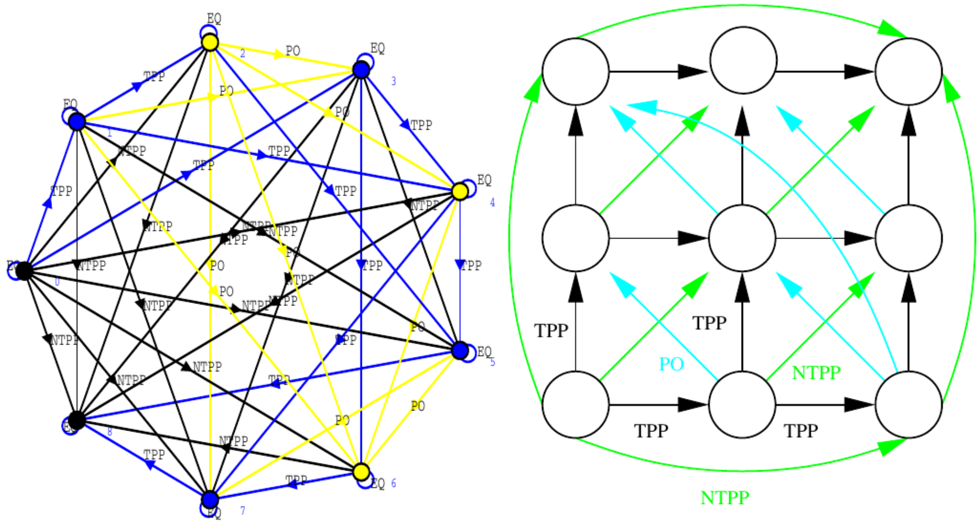

Figure 1: Wessel's original $\mathcal{ALCI}_{\mathrm{RCC8}}$ grid construction, reproduced *directly from report7.pdf (2002/2003)*, Figure 10 (Section 4.4.3): a complete-graph RCC8 "frame" (left) that is isomorphic to a 3×3 grid (right) and extends to $n \times n$. Horizontal and vertical successors are TPP edges (the intended roles R_X, R_Y); PO (cyan) and NTPP (green) are the induced diagonal and long-range relations. Such grid models *exist*—but, as the same report already argued, no concept term can *enforce* them. The construction and the obstruction are the human author's, two decades before this AI-assisted work.

are regular closed subsets of a topological space (typically $\mathbb{R}^n$). They proved:

- L_{RCC8} is **undecidable** (their §5: a linearized NTPP-chain enumeration of tile positions, following Marx–Reynolds, whose faithfulness relies on the geometry of $\mathbb{R}^n$; no two-dimensional grid coincidence is enforced).
- $L_{RCC5}(RS^{\exists})$ is **undecidable** (reduction from $S5^3$ via supremum regions).
- $L_{RCC5}(RS)$ is **open**. Lutz and Wolter explicitly wrote (p. 31): "Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [...] to which the reduction exhibited in Section 8 does not apply."

The logic $\mathcal{ALCI}_{\mathrm{RCC5}}$ is exactly $L_{RCC5}(RS)$: arbitrary complete-graph models with composition-table semantics. The problem has been recognized as open by leading researchers since 2006.

On priority and citation. Two honest caveats, in both directions. First, Wessel proved *neither* decidability nor undecidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ or $\mathcal{ALCI}_{\mathrm{RCC8}}$: report7 constructs the grid models and pins the coincidence obstruction as the precise boundary, but stops there, explicitly leaving both problems open. The contribution was to *locate* the wall, not to breach it. Second, and less comfortably: the Lutz–Wolter paper does not cite report7, although it was a publicly available official technical report of the University of Hamburg (2002/2003) that already developed the $\mathcal{ALCI}_{\mathrm{RCC}}$ description logics, the even-odd counter construction, the explicit two-dimensional TPP/NTPP grid, and the coincidence obstruction as the decisive obstacle. We record this as a matter of *priority*, not of dependence—their undecidability proof takes a genuinely different (topological, linearized) route and in no way relies on report7—but the overlap in ingredients was substantial and the material was findable.

Approach	∀PP.C?	∃DR.D?	Quantify over relations?
Concrete domains, e.g. $\mathcal{ALC}$(RCC8)	no	no	no
Composition-based role boxes ($\mathcal{ALCI}_{\text{RCC5}}$)	**yes**	**yes**	**yes**

Table 3: Concrete domains versus composition-based role boxes. Only the latter can *quantify over* spatial relations, which is exactly what puts $\mathcal{ALCI}_{\text{RCC5}}$ outside the reach of the concrete-domain results.

4.5 Concrete-domain approaches, and why they do not settle it

A natural objection is that decidability of description logics with RCC constraints is surely already known. It is—but for a *different*, expressively incomparable formalism: the *concrete-domain* approach, in which spatial relations constrain *functional attributes* of individuals rather than serving as roles. The distinction is decisive (Table 3). A concrete domain reasons about the spatial *attributes* of elements; $\mathcal{ALCI}_{\text{RCC}}$ instead makes the spatial relations *themselves* the roles, and so permits *quantification over* them—concepts such as $\forall\mathsf{PP}.C$ ("every proper part satisfies C") and $\exists\mathsf{DR}.D$ ("some discrete region satisfies D")—which the concrete-domain setting cannot express.

The known decidability results all sit on the concrete-domain side. Lutz and Miličić [10] showed $\mathcal{ALC}$ with ω-admissible concrete domains (which include RCC5 and RCC8) decidable, without inverse roles; Baader and Rydval [11] revisited and generalized the ω-admissibility conditions (and strengthened the matching undecidability results); Borgwardt, De Bortoli and Koopmann [12] pinned $\mathcal{ALC}(\mathcal{D})$ ontology consistency at ExpTime-complete for ω-admissible $\mathcal{D}$ ($\mathcal{ALC}$ only, no inverse); and Demri and Gu [13] recently extended the constraint-automaton method to *inverse* roles and functional role names with ExpTime membership—the closest published result to $\mathcal{ALCI}_{\text{RCC}}$, yet still within the concrete-domain formalism. None settles $\mathcal{ALCI}_{\text{RCC5}}$ or $\mathcal{ALCI}_{\text{RCC8}}$: the composition-based role box allows quantification over the spatial relations themselves, and the complete-graph semantics is not a concrete-domain constraint. The two lines are complementary—attributes versus relationships—which is why twenty years of progress on the former left the latter open.

5 Decidable fragments of first-order logic, and why none captures it

$\mathcal{ALCI}_{\text{RCC5}}$ satisfiability is, quite literally, a *first-order* satisfiability problem. The five relations are binary predicates; the composition-table semantics is a finite set of first-order axioms T_{RCC5}—exactly one atom on each ordered pair of distinct elements, EQ as identity, converse coherence, and, for every pair of atoms r, s and every $t \notin \mathsf{comp}(r, s)$, the *forbidden triangle* $\forall x\, y\, z\ \neg\big(R_r(x, y) \wedge R_s(y, z) \wedge R_t(x, z)\big)$—while the concept C_0 becomes a sentence $\exists x\,\mathrm{ST}(C_0, x)$ under the standard modal translation. Thus C_0 is satisfiable iff $T_{\text{RCC5}} \cup \{\exists x\,\mathrm{ST}(C_0, x)\}$ has a model. If this theory fell inside a *decidable fragment* of first-order logic we would be done—a natural first thing to try, and, at coarser granularity, exactly how the decidable family members fell: report7 settles $\mathcal{ALCI}_{\text{RCC3}}$ by precisely this route (below). At RCC5 it does not work; the surface reasons vary by fragment, but the deepest one—the one that defeats even the fragments tailor-made for relational theories like this—is the same totality that underlies F6.

Two invariants of $\mathcal{ALCI}_{\text{RCC5}}$ models constrain which fragment could contain it.

- **No finite model property.** An ascending PP-tower ($\exists\mathsf{PPI}.\top \sqcap \forall\mathsf{PPI}.\exists\mathsf{PPI}.\top$, with PP a strict order) is satisfiable only in infinite models. Any fragment with the finite model property is excluded outright.

- **Complete Gaifman graph.** Totality forces every two distinct elements to be related, so the Gaifman graph[1] of every model is *complete*: a PP-chain of length n already has treewidth $n-1$ and clique number n (`wp78`). Any fragment whose decidability rests on models of bounded treewidth—the whole guarded family—is excluded as well.

Against these, the standard fragments fall as follows.

Two variables (FO^2*).* This is how the coarser family members actually fell: report7 proves $\mathcal{ALCI}_{\mathrm{RCC3}}$ (strong EQ) decidable by translating concepts, via the standard translation, into *two-variable* first-order logic with equality [8]—possible there because the RCC3 role theory needs no ternary composition axiom at all (report7's role theory is exactly-one plus converse symmetry, all two-variable; the RCC3 table constrains no triple of distinct elements). At RCC5 the route is excluded twice over: the composition axiom is now genuinely ternary ($\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ constrains distinct triples), and FO^2 has the finite model property [14], which $\mathcal{ALCI}_{\mathrm{RCC5}}$ lacks. Worse, the ternary step is into hostile territory—the three-variable fragment FO^3 is already *undecidable* (it captures the equational theory of relation algebras [15])—so variable-counting alone offers no route: any decidability must come from the *shape* of the axioms, not their arity.

Prefix classes (Bernays–Schönfinkel–Ramsey and relatives). The standard translation of nested modalities produces unbounded $\forall\exists$ alternation, so no bounded quantifier-prefix class applies; these classes also have the finite model property.

The guarded fragment and its relatives. This is the interesting near-miss, and it is worth being precise about where it breaks. The concept part is *guarded* outright—the modal translation is the paradigm guarded formula, each quantifier relativized by its role atom ($\exists\mathsf{PP}.C \rightsquigarrow \exists y(R_{\mathsf{PP}}(x,y) \wedge \cdots)$) [16]. The composition constraint, in forbidden-triangle form, is *clique-guarded*: in $R_r(x,y) \wedge R_s(y,z) \wedge R_t(x,z)$ every pair of variables co-occurs in an atom—exactly a clique guard—so composition-closure lives in Grädel's clique-guarded fragment [17]; converse and mutual-exclusion are guarded too. In other words *almost* the entire theory is clique-guarded. What is not—and cannot be—is **totality**: "every two elements are related" is the archetypal non-guarded assertion (it is not invariant under the guarded bisimulation these fragments are built around). And totality is not a technicality one may drop: it is what forces the complete Gaifman graphs above, which destroy the generalized tree-model property on which the guarded family's decidability rests. The guarded route thus fails for exactly the reason `wp78` records—and note it is *totality*, not the ternary composition axiom, that does the damage; the composition axiom is the tamer half.

Remark 7 (On the two renderings of the composition table)**.** The *natural* first-order rendering of a table cell is not the forbidden triangle but the implication

$$\forall x\, y\, z\, \big(R_r(x,y) \wedge R_s(y,z) \;\rightarrow\; R_{t_1}(x,z) \vee \cdots \vee R_{t_n}(x,z)\big), \qquad \{t_1, \ldots, t_n\} = \mathsf{comp}(r,s),$$

and it is instructive that this form is *less* guarded, not more. Its guard $R_r(x,y) \wedge R_s(y,z)$ is a *path*, not a clique: the pair $\{x,z\}$ the consequent speaks about never co-occurs in a guard atom—indeed for $r = s = t_1 = \mathsf{PP}$ the sentence *is* transitivity, $\forall xyz\,(R_{\mathsf{PP}}(x,y) \wedge R_{\mathsf{PP}}(y,z) \rightarrow R_{\mathsf{PP}}(x,z))$, the textbook non-guardable shape (and the one behind the transitivity undecidability cited below).

[1] The *Gaifman graph* of a relational structure is the undirected graph on its elements in which two distinct elements are joined whenever they occur together in some tuple of some relation—for binary relations, whenever $R(x,y)$ or $R(y,x)$ holds for some R. It is the standard way to speak of the "locality" or "sparsity" of a model: bounded degree, bounded treewidth, and similar tameness conditions are all read off it. Here, because every pair of elements carries an RCC5 atom, this graph is *complete*—maximally non-local.

The contrapositive forbidden triangles are clique-guarded precisely because the offending third edge moves *into the guard*. The two renderings are equivalent only modulo exactly-one: the implication additionally *forces* an atom on (x, z)—a relativized piece of totality—and the negative rewriting is available at all only because the relation alphabet is finite and exhaustive (for a single transitive relation over an open vocabulary no such rewriting exists). Nor is the rewriting a loophole: within the clique-guarded fragment the triangle sentences are *strictly weaker* than transitivity, since a model may leave (x, z) atom-free; the equivalence broker is totality, which is exactly the axiom the fragment cannot have. Nothing downstream is sensitive to the choice—both are finite first-order renderings, interchangeable over T_{RCC5}, so the Π^0_1 placement (§7.5) is unaffected—but the diagnosis is: written with implications, composition *looks* unguarded for a repairable, surface reason; written with triangles, every axiom of T_{RCC5} dresses in guards *except* totality, which is the honest accounting.

Other fragments. The fluted fragment requires predicate arguments to form a suffix of the quantifier sequence; the composition atom $R_t(x, z)$ skips y, so it is not fluted. The unary-negation and guarded-negation fragments are a subtler near-miss: negation of *sentences* is unrestricted there, so the forbidden triangles themselves *are* expressible (each is the negation of an existential-positive sentence)—but both fragments have the finite model property [18], which $\mathcal{ALCI}_{\mathrm{RCC5}}$ lacks, and totality is again out of reach (its inner negations carry two free variables and no atom to guard them). And the one extension that would embrace the vertical structure directly—the guarded fragment *with transitive relations*, since $\mathsf{PP}, \mathsf{PPI}$ are transitive ($\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$)—is *undecidable* once transitivity is used outside guards, as it must be here [19], so even the most sympathetic extension supplies no decision procedure.

The lesson. The surface reasons vary—arity, prefix shape, atom shape, the finite model property—but those are boundary mismatches that fragment engineering could in principle negotiate. The failure that matters is the guarded family's, and there the culprit is *totality*, i.e. the complete Gaifman graph: the one axiom of T_{RCC5} no guard can carry, and the one obstruction no choice of fragment removes, because it is a property of the *models*, not of the axioms' syntax. It is precisely the obstruction F6 names from the width side and `wp78` from the cover side; the FO-fragment view rediscovers it from classical model theory. It also tells us which tools are the wrong ones: guardedness and locality (Gaifman, Hanf) are exactly what totality defeats, so the productive machinery is the totality-tolerant kind—Ehrenfeucht–Fraïssé / back-and-forth and the composition method over bounded *tuple* profiles—which is the "correct room" the keystone analysis independently points to (§10).

6 Why standard undecidability reductions fail

Undecidability proofs for description logics use a variety of techniques: grid encoding ($\mathbb{N} \times \mathbb{N}$ domino tiling, requiring functional roles or number restrictions), reduction from the Post Correspondence Problem (PCP), reduction from context-free grammar (CFG) intersection, or simulation of $S5^3$ via geometric constructions. $\mathcal{ALCI}_{\mathrm{RCC5}}$ lacks the features that each of these techniques exploits: it has no functional roles, no number restrictions, no arbitrary role box, and no geometric coincidence conditions.

6.1 Blocked reduction routes

Table 4 summarizes the blocked routes. The **patchwork property** of RCC5 (Renz & Nebel [2])—local (triple-wise) consistency implies global consistency for atomic networks—actively resists grid

Source Problem	Technique	Missing in $\mathcal{ALCI}_{\mathrm{RCC5}}$
$\mathbb{N} \times \mathbb{N}$ domino	Graded modalities + transitivity	Number restrictions
$\mathcal{ALC}_{\mathcal{RA}}$ [6]	Post Correspondence Problem	Arbitrary role axioms
$\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$ [3]	Non-empty CFG intersection	Non-disjoint roles
$\mathcal{ALCI}_{\mathrm{RCC8}}$ (Wessel)	Grid via $\mathsf{TPP}/\mathsf{NTPP}$	$\mathsf{TPP}/\mathsf{NTPP}$ distinction
L_{RCC8} (Lutz–Wolter)	Linearized topological enumeration	Topological rigidity
$L_{RCC5}(RS^{\exists})$	$S5^3$ via supremum regions	Supremum closure

Table 4: Why standard undecidability reductions fail for $\mathcal{ALCI}_{\mathrm{RCC5}}$.

encoding, since tiling reductions require rigid global constraints that go beyond local consistency.

6.2 A concrete domino attempt in $\mathcal{ALCI}_{\mathrm{RCC5}}$

Can we encode a domino-like grid using PP/PPI chains in RCC5? Consider the concept:

$$\begin{aligned} X \; = \; & \exists\mathsf{PPI}.\exists\mathsf{PPI}.C \;\sqcap\; \exists\mathsf{PPI}.\exists\mathsf{PPI}.D \\ & \sqcap\; \forall\mathsf{PPI}.\forall\mathsf{PPI}.(\forall\mathsf{PO}.(\neg C \sqcap \neg D) \sqcap \forall\mathsf{DR}.(\neg C \sqcap \neg D) \\ & \qquad\qquad \sqcap\; \forall\mathsf{PPI}.(\neg C \sqcap \neg D) \sqcap \forall\mathsf{PP}.(\neg C \sqcap \neg D)) \\ & \sqcap\; \forall\mathsf{PPI}.X \end{aligned}$$

The intent: each X-node has two PPI-grandchildren colored C and D (the "square"), and the universals isolate the colored nodes from all non-EQ neighbors. The recursive conjunct $\forall\mathsf{PPI}.X$ forces infinite repetition.[2]

One level is satisfiable. Without the recursive conjunct, X has a 3-element model: $z \;\mathsf{PP}\; y \;\mathsf{PP}\; x$, with $z \in C \sqcap D$. The two grandchildren are forced to be *the same element* (EQ collapse): if $z_1 \neq z_2$, they would be related by some $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, and the universals would require $z_2 \notin D$ (firing z_1's $\forall R.\neg C \sqcap \neg D$). So $z_1 = z_2 \in C \sqcap D$, and the universals are vacuously satisfied (the only neighbors of z are PP-predecessors, not PO/DR/PPI-neighbors).

Two levels is unsatisfiable. Adding $\forall\mathsf{PPI}.X$: the PPI-child y must itself satisfy X (without recursion), producing a new $C \sqcap D$-labeled element w as a PPI-child of z. But z is a PPI-grandchild of x, and x's universals fire: z's PPI-neighbor w must satisfy $\neg C \sqcap \neg D$. Yet $w \in C \sqcap D$. **Contradiction.**

This was verified computationally: a simplified version $X_1 = \exists\mathsf{PPI}.\exists\mathsf{PPI}.C \sqcap \forall\mathsf{PPI}^3.\neg C$ is SAT; $X_2 = X_1 \sqcap \forall\mathsf{PPI}.X_1$ is UNSAT. The domino encoding collapses at depth 2 because the isolation constraints from the grandparent level kill the colored elements created by the recursive unfolding.

Remark 8. The failure is not accidental. In $\mathcal{ALCI}_{\mathrm{RCC5}}$, the only way to create "functional" successors is via PP/PPI chains, but $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ makes these chains *transitive*—grandparent universals automatically propagate to grandchildren, preventing the positional diversity needed for grid encoding. In $\mathcal{ALCI}_{\mathrm{RCC8}}$, the $\mathsf{TPP}/\mathsf{NTPP}$ distinction breaks this transitivity (a direct TPPI-child is distinguishable from indirect descendants), which is why Wessel's even-odd chain works there but has no RCC5 analogue.

[2] The recursive occurrence of X abbreviates a TBox axiom $X \sqsubseteq F(X)$; no fixpoint constructor is needed. Since the five relations partition the square of the domain, the *universal modality* $[U]D := \forall\mathsf{EQ}.D \sqcap \forall\mathsf{PP}.D \sqcap \forall\mathsf{PPI}.D \sqcap \forall\mathsf{PO}.D \sqcap \forall\mathsf{DR}.D$ is definable, and a finite TBox internalizes: C is satisfiable w.r.t. $\mathcal{T}$ iff $C \sqcap [U]G_{\mathcal{T}}$ is, where $G_{\mathcal{T}}$ conjoins $\mathrm{nnf}(\neg C_i) \sqcup D_i$ over the axioms $C_i \sqsubseteq D_i$ of $\mathcal{T}$. The internalization uses $\forall\mathsf{PO}$, so it does *not* preserve the $\forall\mathsf{PO}$-free fragment of Theorem 10.

6.3 A concrete domino attempt in $\mathcal{ALCI}_{\mathrm{RCC8}}$

The remark above suggests the $\mathsf{TPPI}/\mathsf{NTPPI}$ distinction in $\mathcal{ALCI}_{\mathrm{RCC8}}$ might rescue the construction: a direct successor (TPPI) and an indirect one (NTPPI) are genuinely distinguishable, so perhaps the level-2 collapse can be avoided. We attempt the analogue and show that while it escapes the RCC5 failure mode, it fails for a different reason—Wessel's *coincidence obstruction* [8].

The schema. Let Corner be the concept that isolates a node from all non-EQ neighbors:

$$\mathsf{Corner} \;=\; \bigwedge_{R\in\{\mathsf{DC},\mathsf{EC},\mathsf{PO},\mathsf{TPP},\mathsf{NTPP},\mathsf{TPPI},\mathsf{NTPPI}\}} \forall R.(\neg C \sqcap \neg D).$$

The isolation is placed *locally on the colored grandchildren* rather than as an ancestral $\forall\mathsf{TPPI}.\forall\mathsf{TPPI}$ universal, so it cannot propagate transitively through the recursion:

$$\begin{aligned} X \;=\; & \exists\mathsf{TPPI}.\exists\mathsf{TPPI}.(C \sqcap \mathsf{Corner}) \;\sqcap\; \exists\mathsf{TPPI}.\exists\mathsf{TPPI}.(D \sqcap \mathsf{Corner}) \\ & \sqcap\; \forall\mathsf{TPPI}.(\neg C \sqcap \neg D) \;\sqcap\; \forall\mathsf{TPPI}.X. \end{aligned}$$

The third conjunct $\forall\mathsf{TPPI}.(\neg C \sqcap \neg D)$ forces the corner relation $x \to z$ to be NTPPI (not TPPI), using $\mathsf{comp}(\mathsf{TPPI},\mathsf{TPPI}) = \{\mathsf{TPPI},\mathsf{NTPPI}\}$.

Level 1 is satisfiable, with a real direct/indirect distinction. Without recursion, x has TPPI-children y_1, y_2 (the two "direct" positions) and a single corner z reached by TPPI-TPPI chains from both y_1 and y_2. The EQ-collapse at level 2 works exactly as in RCC5: $z_1 \in C \sqcap \mathsf{Corner}$ and $z_2 \in D \sqcap \mathsf{Corner}$ are mutual non-EQ neighbors only if Corner is violated, so under strong-EQ semantics $z_1 = z_2 = z$. Crucially, $z \neq y_1, y_2$ in general: z is NTPPI of x while y_1, y_2 are TPPI—the direct/indirect distinction survives, which is a genuine RCC8 refinement over the RCC5 attempt of §6.2.

Level 2 collapses via the coincidence obstruction. Adding $\forall\mathsf{TPPI}.X$: the TPPI-child y of x satisfies X, producing its own corner $w \in C \sqcap D \sqcap \mathsf{Corner}$ at NTPPI-distance from y, and hence at NTPPI-distance from x via $\mathsf{comp}(\mathsf{TPPI},\mathsf{NTPPI}) = \{\mathsf{NTPPI}\}$. Now x's own corner z and y's corner w are distinct grid positions in the intended reading—but as elements of the model they are related by some RCC8 base relation. Since z carries Corner, every non-EQ neighbor of z lies in $\neg C \sqcap \neg D$. But $w \in C \sqcap D$. Therefore $z = w$ under strong-EQ. **All corners across the grid coincide at a single point.** Figure 2 isolates this local "merge attempt" on two successors.

Why the $\mathsf{TPPI}/\mathsf{NTPPI}$ trick fails. In the RCC5 attempt the collapse is a *transitive-propagation* failure: the grandparent universal fires on great-grandchildren because $\mathsf{comp}(\mathsf{PPI},\mathsf{PPI}) = \{\mathsf{PPI}\}$. Moving the isolation to a local Corner marker and using the non-transitive piece $\mathsf{comp}(\mathsf{TPPI},\mathsf{TPPI}) = \{\mathsf{TPPI},\mathsf{NTPPI}\}$ eliminates that specific pathway—the universal at the grandparent no longer reaches deeper nodes. But a *different* obstruction takes over: the composition table permits any of $\{\mathsf{PO},\mathsf{EQ},\mathsf{TPP},\mathsf{NTPP},\mathsf{TPPI},\mathsf{NTPPI}\}$ between two corners at different intended grid coordinates, and nothing in the concept language rules out EQ. The Corner markers themselves then force the EQ-merge. This is exactly the obstruction Wessel identified in report7.pdf, Figure 11 (cf. §4.3): the composition table does not carry the rigidity needed to keep distinct grid cells distinct.

Remark 9. The failure mode is structurally different from the RCC5 one but traces back to the same root cause. Lutz–Wolter's L_{RCC8} reduction *circumvents* the coincidence obstruction rather than enforcing it: they linearize the plane into a single NTPP-chain (the Marx–Reynolds enumeration) and recover the vertical direction from the enumeration, so no "north-of-east = east-of-north"

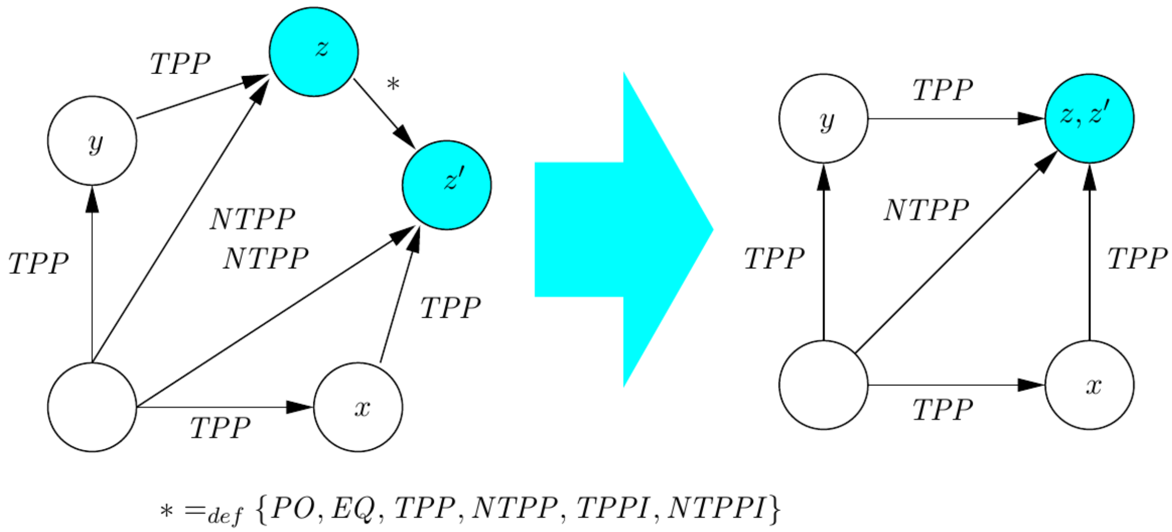

Figure 2: Wessel's original statement of the coincidence obstruction, reproduced *directly from report7.pdf (2002/2003)*, Figure 11 ("Tightening the grid?"). The two successors z (reached east-then-north) and z' (north-then-east) ought to be the *same* grid corner, but the composition table permits the edge $z * z'$ to be any of $* =_{\text{def}} \{\mathsf{PO}, \mathsf{EQ}, \mathsf{TPP}, \mathsf{NTPP}, \mathsf{TPPI}, \mathsf{NTPPI}\}$ (the legend below the graphs), so nothing *forces* the merge $z = z'$ (right). Read in the present notation: leave it unforced and the two branches never join (no grid); force it with the Corner isolation markers of the construction above and *every* corner collapses to one point. This "merge the wrong successors" problem was identified by the human author two decades before this AI-assisted work.

coincidence has to be forced; the geometry of $\mathbb{R}^n$ only has to keep that linear encoding faithful [9]. No such rigidity is available in the abstract RCC8 semantics of $\mathcal{ALCI}_{\text{RCC8}}$: a pure composition table allows corner points to merge freely, and the $\mathsf{TPPI}/\mathsf{NTPPI}$ distinction—while useful locally for a direct/indirect separation at level 2—cannot be chained to enforce an unbounded $\mathbb{N} \times \mathbb{N}$ grid. This matches the open-problem status of $\mathcal{ALCI}_{\text{RCC8}}$ decidability noted in [9] and further discussed in [8].

7 Overview of approaches, by technique

We organize the attack by *technique*, in roughly the order tried, not by round or date. Each subsection follows the same shape: the high-level idea; the concrete techniques investigated; where and why they fail (with the examples that made the failure clear); pointers to the papers, probes, and reasoners; and the lessons.

7.1 Naive tableaux and blocking

The idea, and why it should work. The default decision method for a description logic is a *tableau*: to test whether C_0 is satisfiable, try to build a model top-down. Start with one individual required to satisfy C_0; whenever a node must satisfy $\exists R.D$, create a fresh R-successor satisfying D; whenever it satisfies $\forall R.D$, propagate D to all its R-successors; close a branch on a contradiction. Left alone this may never terminate—$\exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$ builds an infinite PP-chain—so one adds *blocking*: if a new node's "type" (the set of concepts it must satisfy) repeats that of an ancestor, we stop expanding it and reuse the ancestor's already-built subtree, folding the infinite chain into a finite loop. For ordinary modal and description logics this is sound and complete, and it is the first thing one tries here. Our most refined implementation is the *cover-tree tableau* (§8), which blocks on *exact occurrences.*

Why the total-relation setting breaks it. In $\mathcal{ALCI}_{\mathrm{RCC5}}$ *every* pair of individuals carries a relation, so a model is a complete graph, not a tree. This single fact defeats ordinary blocking in three compounding ways, each of which we hit in practice.

1. **Composition forces edges *into* already-built nodes.** Creating a new node z does not just attach it to its parent; by composition it forces relations between z and *every* existing node. From $x\,\mathsf{PP}\,y$ and $y\,\mathsf{DR}\,z$ the table forces $x\,\mathsf{DR}\,z$ (Proposition 1). So the "tree" is a fiction: the real object is the whole complete graph, and a node's obligations depend on its relations to the entire context, not on its concept type alone.

2. **Dormant universals reawaken blocked nodes.** A restriction $\forall R.D$ lies inert at a node until an R-edge from it appears. But a *forced* composition edge (point 1) can create such an edge *after* the node was blocked, firing the dormant universal, adding D to a node downstream, and possibly *unblocking* it. The cascade of re-qualifications need not terminate, and—worse—may be *unsound* if ignored: the block asserted a repetition that the later edge invalidates.

3. **A fresh witness's relations are forced by the *joint* closure, and can collide with a universal.** Adding d does not leave its relations to old nodes free: they are pinned by *all* triangles at once, and (Remark 2) the intersection of several can fix even a horizontal edge. Take old $a\,\mathsf{DR}\,b$, $a\,\mathsf{PPI}\,c$, $b\,\mathsf{DR}\,c$ and a request $b\,\mathsf{PO}\,d$, $c\,\mathsf{PO}\,d$: this is consistent, but it *forces* $a\,\mathsf{PO}\,d$—from $\mathsf{comp}(\mathsf{DR},\mathsf{PO})\cap\mathsf{comp}(\mathsf{PPI},\mathsf{PO}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PP}\}\cap\{\mathsf{PPI},\mathsf{PO}\} = \{\mathsf{PO}\}$, an edge a never chose. Now let a carry $\forall\mathsf{PO}.\neg A$ while d must satisfy A: the *forced* $a\,\mathsf{PO}\,d$ fires the universal and closes the branch. That is the real hazard—a value fixed by joint closure, not by any single step, silently contradicting a distant universal (the diagnostic C_{force} below is exactly this shape)—and it is invisible to any check that reads edges one path at a time.

The ladder of blocking conditions we tried. Each rung tracks more relational context, and each is defeated by the phenomena above. The base choice is *where* one may block: the obvious strengthening of ancestor blocking is *anywhere blocking*—block a node against *any* earlier node in the tableau, not only its ancestors, the more aggressive termination device standard for logics with inverse roles. This helps termination but not soundness: it inherits every failure below, since it still compares nodes by a *local* summary while the constraints are global. On top of the where-question we layered progressively richer *what*-summaries. *Subset/type* blocking (repeat of the concept set) ignores relations entirely and is unsound at once. *Triangle* blocking tracks, for each node, the composition-*triangles* it participates in—the relation triples $(\rho(x,y),\rho(y,z),\rho(x,z))$—so that composition constraints are visible; but a triangle is a three-point view, and forced edges from a fourth point (via inverse roles) escape it (`triangle_blocking_ALCIRCC5`, `src/triangle_*.py`). *Quadruple* types lift this to four points to capture those cross-edges (`decidability_via_quadru-ples_ALCIRCC5`, `src/quadruple_type_check.py`); but the required arity is not bounded—a forced constraint can span a chain of any length. *Profile* blocking records a node's type together with the multiset of its relations to a bounded context (`src/profile_blocking_*.py`); but the context that actually determines future obligations is not bounded a priori—this is F6 in disguise. *Double* blocking (the $\mathcal{SHIQ}$ device: block only when a node *and its predecessor* jointly repeat an ancestor pair, which is what one needs for inverse roles) addresses point 2's predecessor sensitivity but not the forced-edge and extension problems (`src/double_blocking_test.py`).

The examples that settle it. Two families are diagnostic and recur throughout the project. The forced-composition UNSAT concepts, e.g. $C_{\text{force}} = \exists\mathsf{PP}.(\exists\mathsf{DR}.A) \sqcap \forall\mathsf{DR}.\neg A$: here $x\,\mathsf{PP}\,y$ and $y\,\mathsf{DR}\,z$ force $x\,\mathsf{DR}\,z$, so $\forall\mathsf{DR}.\neg A$ kills the required $z:A$; a procedure that ignores forced edges wrongly *accepts* it. Dually, the strong-EQ PO-*loop* $C \sqcap \exists\mathsf{PO}.\exists\mathsf{PO}.C \sqcap \forall\{\mathsf{PO},\mathsf{DR},\mathsf{PP},\mathsf{PPI}\}.\neg C$ is satisfiable *only* via a two-element model closed into a symmetric PO-loop under EQ-identity; a procedure that distinguishes occurrences by type alone wrongly *rejects* it (this is precisely the blind spot of the quasimodel oracle, §7.2). Further diagnostics—`inter_witness_counterexample_ALCIRCC5`, `typed_patchwork_counterexample_ALCIRCC5`—exercise the same seams.

Lesson. In a total-relation logic, sound termination cannot be had by *local* blocking, because the relational completion is *global*: what may be reused depends on the whole graph, and forced edges plus dormant universals make "the whole graph" irreducible to any fixed-arity local view. The way out is to stop trying to block a complete graph and instead *separate* the tree skeleton from the global relational completion, controlling the latter by the RCC5 patchwork property. That is the split-forest idea (§7.3).

7.2 Finite-model and reduction attempts

The idea. If one could force models to be finite, or finitely presented, one could enumerate candidates and check them. Four concrete lines: build a *quasimodel* (a set of types with a consistency relation) and eliminate types that cannot be realized; collapse infinite PP/PPI chains into finite *clusters or loops*; take a *two-tier quotient* (one "kernel" per recurrent chain descriptor, plus finitely many designated witnesses); or *encode* satisfiability into monadic second-order logic (MSO) over trees and invoke Rabin's theorem.

Where they fail. The quasimodel / type-elimination procedure (`decidability_ALCIRCC5`, `src/alcircc5_reasoner.py`) is *retracted*: its elimination rule for PO-demands ($Q3$) is *anti-monotone*—eliminating one type can invalidate the witness that justified another—so the fixpoint over-eliminates and the procedure is *incomplete* (though not unsound). Concretely it is wrong on the PO-loop above: it reports UNSAT for a satisfiable concept. The finite-cluster and MSO routes both stumble on the same rock: collapsing a PP-chain is fine (Lemma 4), but the *horizontal* width across clusters is not bounded, and MSO cannot define an unbounded horizontal matching—so a correct encoding again presupposes a width bound. Every "make it finite" variant reduces to producing a computable live-width bound, i.e. to F6 (Definition 5).

The win. One fragment escapes unconditionally. In the two-tier quotient, PO is the *only* relation whose recurrence along a chain needs an "all-phase" safety condition—because, unlike DR/PP/PPI, a PO-witness at one chain position need not remain a PO-witness at another (the compositions $\mathsf{comp}(\mathsf{PP},\mathsf{PO})$ and $\mathsf{comp}(\mathsf{PPI},\mathsf{PO})$ are not singletons). If the concept contains *no* $\forall\mathsf{PO}$ subformula, that safety condition is vacuous, the quotient is finite, and the width is bounded by a computable function of the concept.

Theorem 10 (Decidable fragment). *Concept satisfiability for the* $\forall\mathsf{PO}$*-free fragment of* $\mathcal{ALCI}_{\text{RCC5}}$ *—concepts with no* $\forall\mathsf{PO}$ *subformula, but retaining* $\exists\mathsf{PO}$, $\forall\mathsf{DR}$, $\exists\mathsf{DR}$, *and all vertical modalities—is* *decidable, with a computable live-width bound* $K(C_0) = (1+m)^2\, 2^{2^m}$, $m = |\mathsf{cl}(C_0)|$. (`two_tier_-quotient_ALCIRCC5`, `po_free_fragment_ALCIRCC5`; theorem-level, resting on the two-tier quotient lemmas.)

The fragment is genuinely expressive—one may still say "overlaps some A" ($\exists\mathsf{PO}.A$), "all disjoint things are B" ($\forall\mathsf{DR}.B$), and impose arbitrary part-of nesting—so this is the project's strongest unconditional $\mathcal{ALCI}_{\mathrm{RCC5}}$ result.

Re-examined, and its soundness core certified (2026-07-18). A cold review (§9) rightly flagged that the rationale above leans on the "horizontal freedom" intuition that Remark 2 corrects. We therefore re-examined the fragment; it holds. The soundness crux—the *chain-unfolding lift* (a recurrent chain's kernel, unfolded under a constant interface, stays a valid RCC5 frame)—is now *kernel-checked* in Lean 4 (`formal/POFreeLift.lean`: `lift_cc`, `unf_is_frame`; axioms `propext`/`Quot.sound`, zero `sorry`), and it enforces *full* composition-consistency (every triangle closed), which *subsumes* multi-path PO forcing—so the construction does *not* assume PO edges are free, closing the review's concern. Two independent decision procedures moreover agree on every $\forall$PO-free concept tested (`wp86`, `wp87`: 244 random + curated, zero mismatches; the quasimodel oracle's lone blind spot requires $\forall$PO and so cannot arise here). A byproduct worth noting: the abstract-semantics core needs *no* patchwork or compactness—a composition-closed, converse-coherent, strong-EQ network already *is* a model, so `unf_is_frame` delivers an actual (infinite) frame outright. What remains for full certification is the model-of-C_0 layer (Hintikka demand/universal satisfaction) and the completeness extraction carrying the $K(C_0)$ bound. So Theorem 10 now stands as *strongly supported with its soundness core machine-certified*—firmer than before the review, though not yet end-to-end formalized.

Lesson. Finiteness is the wrong target: models are legitimately infinite (unbounded PP-towers are satisfiable). The right target is a finite *presentation* of a possibly-infinite model.

7.3 The split-forest idea

Idea (the human author's, on a whiteboard). The split-forest decomposition was the human author's contribution, sketched on a whiteboard (Figure 3) and handed to the AI assistants to make precise. The insight: even though a model is a complete graph, one can *present* it as a *forest of trees*—one tree per connected component of the vertical (PP) order—by *splitting* each node that is a common part of several superparts into separate *occurrence copies* (which are *not* EQ-related model elements—under strong EQ that would mean identity—but occurrences of an occurrence cover, re-identified afterwards by an explicit quotient), so the tree skeleton becomes explicit, and then *completing* the missing (horizontal) edges by the RCC5 patchwork property. Tree-like presentations of infinite models are the classical gateway to decidability: once a logic has a tree-model property, automata or type-elimination usually finish the job. The bet was that this setting, despite its complete relational graph, still admits such a presentation.

The normal form (Theorem A). Made precise, the construction is a chain of thinning and splitting operations that every satisfiable concept survives: *witness thinning* (keep one witness per demand) → *generated containment covers* (organize witnesses by the vertical order) → *split copies* for incomparable superparts → *side-interface mosaics* (the horizontal relations at each interface, completed by patchwork) → a *typed*-EQ *quotient* (re-identify the split mates by exact occurrence).

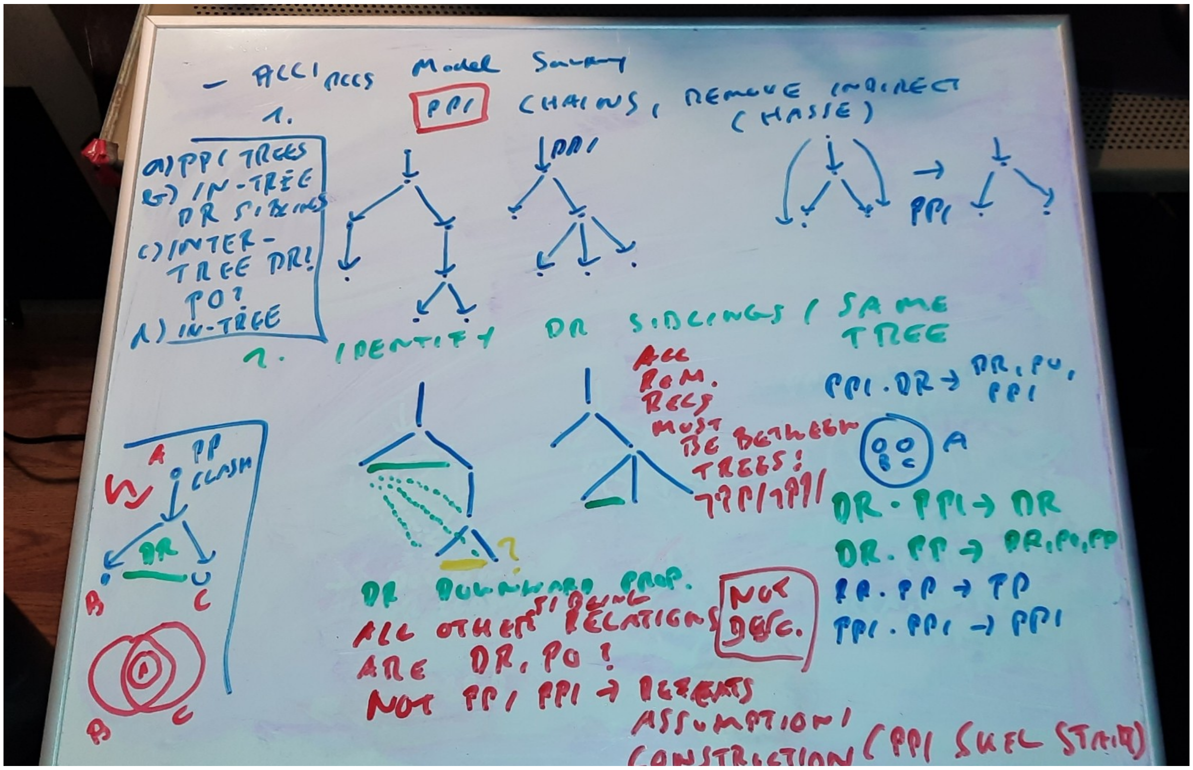

Figure 3: The original whiteboard sketch of the split-forest model decomposition (M. Wessel, April 2026): the intuition that a complete-graph RCC5 model can be presented as a forest of trees by splitting join nodes into occurrence copies (re-identified by a quotient) and completing the relational graph by patchwork. This idea—the human author's—is the semantic foundation that every subsequent proof route consumes, and the one component no cold review ever broke.

The output is a witness-generated, tree-like split-forest model. This normal form is the one component found sound by *every* cold review across the project—the semantic foundation everything downstream *consumes* rather than replaces (`trees/`, `cover_tree_tableau_ALCIRCC5`).

Vertical skeleton, horizontal completion. The presentation enforces a strict division of labour between the two axes, and that is what makes it a *forest* rather than an arbitrary graph. Every *tree edge* is vertical: a parent-to-child link is a PP/PPI (part-of) step, so each tree is literally a branch of the containment order. Every *non-tree edge*—between siblings, cousins, or nodes in different trees—is *horizontal* (DR or PO). A genuine vertical relation that happens to hold between two nodes not yet on one ancestor line is never left lying across the forest: it is *re-oriented* into the tree (the smaller region is made a descendant of the larger), so that once the splitting is finished the only edges the trees do *not* already display are horizontal ones. This is exactly why the completion problem is a purely *horizontal* one—and why PP/PPI never appear as edges to be guessed at an interface.

Half-determinism (Proposition 3) then does most of the completion for free. Once a DR edge is placed between two subtrees, *every* pair of descendants across them is forced to DR—all shadows, none of them live—so a DR crowd of any width is generated by a few high DR edges and pushed downward deterministically, costing a finite presentation almost nothing. What resists this is precisely the residue: PO (never forced by a single step, and not downward-propagating, Proposition 1) and the *upward* direction of disjointness (a superpart of a disjoint region is unconstrained). The genuine branching a finite presentation must record—the live width (Definition 5)—is this horizontal residue and nothing else, which is why F6 is a statement about horizontal crowds alone.

Why it does not, by itself, decide. The split forest gives the *shape* of models; it does not give a terminating *search*. Turning "a tree-like model of this shape exists" into "an algorithm can decide whether one exists" requires a *finite abstraction* of the (still infinite) split forest—and that finite-abstraction problem is exactly where the next two families live, and where the difficulty concentrates.

7.4 The no-automata certificate, and the automata route

The hand certificate (rounds 2–10). The first attempt at a finite abstraction was a hand-built *certificate*: a finite object whose validity (checked by local conditions on converse, composition, universal safety, request discharge, and residual frontiers) was meant to be equivalent to satisfiability (`gpt5.5_round2/`, `gpt5.5_round4/`). Across ten rounds it worked only by importing, piece by piece, machinery that is really *automaton-shaped*: the conditions that keep a certificate finite while a model is infinite are "request-closed cycles" (a demand and its witness recur), which behave exactly like Büchi/parity acceptance, and "finite product-state exhaustion," which behaves like reachability in a product automaton. Cold reviews found a succession of gaps—an infinite ancestor PPI-tower the construction missed (D-1), then its dual descendant PP-tower, then boundary non-determinacy, then (twice) that "finite checkability" was asserted rather than proved. The honest reading is that the hand certificate was an automaton written informally; so the project made the automaton explicit.

The parity-automaton route (rounds 11–12). This organizes the argument into three theorems. **Theorem A**: the split-forest normal form (above). **Theorem B**: a *finite abstraction* (frontier/pair/triple summaries of the split forest) captures saturated split forests up to satisfiability of all closure formulas. **Theorem C**: a two-way alternating parity *tree* automaton accepts exactly the valid abstractions, and its emptiness is decidable (Vardi). The automaton *consumes* the split forest; it does not replace it. Theorem B is the single load-bearing keystone. Cold reviews (the 5th and 6th) found its global-coherence step *assumed, not proved*: it derived a globally consistent RCC5 labelling of an infinite tree from purely local checks, via a "confluence law" that was never verified. Round 12 rebuilt the abstraction on *patchwork bags* (finite complete atomic networks that agree on overlaps), so global consistency follows from the citable RCC5 *patchwork property* (Renz–Nebel), and the countable completion by a compactness argument over the finite relation alphabet; eventualities are decided by two-way emptiness. A 7th (hot) review then found a residual *Shadow Amalgamation* gap in the truth lemma: the declared shadow edges were asserted to be jointly realizable with the bag networks, which is false without a further discipline (`automata_route_repairs/`, `cold_review_theoremB/`, probes `wp13–wp17`).

Why we abandoned the tree automaton. A cold review of the decision layer exposed a structural problem. The automaton reads a labelled tree and decides accept or reject. But the tree does *not* carry all the data that makes an abstraction valid: it omits the relations between each fresh witness and the "sources" of the universal restrictions ($\forall R.D$) that act on it. A tree is valid exactly when *some* consistent choice of those missing relations exists—so the automaton would have to solve a *guessing* problem ("does some completion of the missing data work?"), not merely read off the tree. The trouble is how an alternating tree automaton works: it explores the tree in many independent parallel copies, and each copy would guess the missing relations *locally*, with nothing forcing different copies to guess *consistently*. One can therefore build a tree in which two copies, running over two different witnesses, each privately guess relations that are locally fine but mutually incompatible, so every copy accepts—even though the true relation between those two witnesses is forced by composition to be impossible ($\mathsf{comp}(\mathsf{PP},\mathsf{PP}) \cap \mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{PP}\} \cap \{\mathsf{DR}\} = \emptyset$). The automaton accepts an unsatisfiable input: acceptance without validity. The three natural repairs (annotate the tape—but the enrichment is unboundedly large; quotient the profiles—provably lossy; guess canonically—needs an unproven uniformization) are each blocked by facts the construction itself establishes. So the tree automaton was replaced by a mosaic / type-elimination procedure directly on the infinitary abstraction (§7.5). The keystone survived the change untouched: Theorem B's finite-abstraction adequacy is F6 in automaton clothing.

7.5 The Lean line and F6

The mathematical basis, and the pivot that produced it. The Lean development is *not* an automaton—and that is the point of a deliberate pivot. The route of §7.4 tried to decide satisfiability with a two-way parity *tree automaton* (Theorem C); the cold review that found "acceptance without validity" showed that a tree automaton cannot work here, because its independent parallel run copies cannot be forced to guess the missing horizontal relations *consistently*. Rather than patch the automaton, the project moved the decision *off* it and back onto the semantic objects the split forest already supplies. What resulted is a *cluster-quasimodel* construction in the *mosaic / type-elimination* tradition of modal and description logic: a finite catalogue of interface *clusters* (mosaics), a consistency relation for gluing them (controlled by the RCC5 patchwork property), and a *Hintikka truth lemma* for correctness—the same tableau-family argument as §7.1, but run on a *finite abstraction* rather than on an unbounded tableau. Unlike the retracted early quasimodel (§7.2), the procedure *enumerates and finitely checks* catalogues rather than running an elimination fixpoint, so it does not inherit that route's anti-monotone incompleteness. The split-forest normal form (Theorem A) is the shared semantic foundation that both the automata route and this one *consume*.

What makes the automaton *unnecessary*, rather than merely replaced, is *one-step demand fulfilment*. An automaton earns its keep on *eventualities*: promises—"some part satisfies D"—that may be discharged arbitrarily far away, and whose fulfilment along an infinite branch is exactly what a Büchi/parity acceptance condition checks. Here every $\exists R.D$ is instead fulfilled by a witness *already present in the catalogue*, so there are no promissory obligations, no eventualities, and hence no acceptance condition to decide—the one task only an automaton did well. Decidability then reduces to enumerating finite catalogues and checking each by a finite composition test, with no automaton in the loop. This is the construction that was formalized in Lean over rounds 19–30.

Lean, and how we use it. Lean 4 is a proof assistant whose kernel type-checks every inference and refuses a proof with a gap. In a project where prose proofs repeatedly *assumed* the hard step, the kernel is a ground-truth oracle that cannot be talked past. We moved the load-bearing soundness argument into *Lean 4 core*—no `mathlib`, no external libraries, no build system: every object is built from first principles, which keeps the trusted base small and the axiom footprint auditable (only `propext` and `Quot.sound`, plus `native_decide` in isolated witness lemmas). The encoding is deliberately direct, so that the Lean statements are legible as the mathematics they formalize: the five RCC5 atoms are an inductive type with the composition table and converse as total functions; concepts are an inductive type in negation normal form (inverse roles absorbed via the converse); an *interpretation* is a domain predicate together with a *total* relation function $\rho : \Delta \times \Delta \to \{\mathsf{EQ},\mathsf{PP},\mathsf{PPI},\mathsf{PO},\mathsf{DR}\}$ and an atomic valuation; satisfaction is defined by structural recursion (a role R is read directly as $\rho\, x\, y = R$); and the RCC5 frame conditions (reflexive-EQ, strong-EQ, converse-coherence, composition closure) are a `structure` over ρ. On top of this foundation sit Hintikka labellings and the certificate objects described below. The development is `formal/Round19Transport.lean` ($\approx$5,300 lines, zero `sorry`), with the full artifact-by-artifact history in `LEAN.md`.

What a certificate is. Concretely, a *certificate* is an explicit finite object whose validity is checkable by finite composition tests. It is worth spelling out what one looks like, since "catalogue" and "steering" recur throughout. A certificate has two parts.

- A *catalogue*—the finite blueprint. It consists of: a *core network* (a fixed finite complete atomic RCC5 network—the part of the model that does not repeat); a finite set of *attachment templates* (each template is one recipe for gluing a fresh cluster of witnesses onto an existing interface—which old occurrences the new ones attach to, in a *gluing normal form* that keeps the shared part canonical); and a *steering function*, indexed by template, that fixes the *horizontal* relations from the old occurrences to each fresh one. The steering function is the one genuine degree of freedom, because those are exactly the relations composition does not determine (Proposition 1); the vertical edges are already carried by the tree.
- A *plan*: a finite recipe naming which template to apply at which interface, in order. *Unfolding* the plan—applying templates step by step, possibly reusing one template unboundedly often to grow an infinite PP-tower—produces the (possibly infinite) split-forest presentation of §7.3.

A certificate is *sound* if a finite check on the catalogue alone—the "S-condition," independent of how long any plan runs—guarantees that *every* unfolding is a proper, composition-closed RCC5 network; this is the step certified in Lean (below). Its *live width* is the largest number of live (non-shadow) occurrences that must coexist at any one interface during unfolding—the quantity F6 must bound. Demand fulfilment is *one-step*: every $\exists R.D$ is satisfied by a witness actually present in the catalogue, so there are no promissory obligations and hence no parity/eventuality condition of the kind the tree automaton needed.

It was on this construction, finally explicit and free of hidden choices, that the entire remaining difficulty localized to two named properties: **F6** (Definition 5, a computable live-width bound) and a uniformization property **W2′**.

Definition 11 (Uniformization, W2′)**.** Extraction reads a finite certificate off a model by summarizing each interface into a bounded *state* (a type, plus the relations to the finitely many recorded anchors). W2′ asks that this summary be *faithful*: whenever two old occurrences carry the *same* state, a fresh witness can be glued onto both in the *same* way—so a glue step depends only on the state, and the catalogue's finitely many states really do capture every gluing. If W2′ fails, two occurrences that look identical to the catalogue may nonetheless demand different completions, so the finite catalogue *under*-represents the model and the checker can *reject a satisfiable concept*. The error is one-directional: a failure of W2′ can only cause over-rejection, never acceptance of an unsatisfiable concept—so W2′ bears on completeness alone, never on soundness.

What is machine-checked (rounds 19–30). The soundness pipeline is kernel-checked end to end: certificate syntax $\to$ an S-condition on the certificate $\to$ a proper atomic RCC5 frame $\to$ the logic layer (a truth lemma, with one-step fulfilment) $\to$ an actual RCC5 model of C_0. So is the *faithfulness* of the finite (Hintikka) abstraction: satisfiability and abstraction-realizability coincide. On top of these sits a fixed, non-oracular Boolean checker of first-order finite certificate data, proved sound, from which decidability follows by finite search—modulo exactly one premise, F6 $\wedge$ W2′.

Theorem 12 (Conditional decidability)**.** *If F6 holds—the live width of models of every satisfiable C_0 is bounded by a computable function of C_0—then concept satisfiability for $\mathcal{ALCI}_{\mathrm{RCC5}}$ is decidable. The* soundness *half of this reduction (a valid finite certificate provably unfolds to a genuine RCC5 model of C_0) and the faithfulness of the Hintikka abstraction are* machine-checked in Lean 4 *(zero* `sorry`*).* (The uniformization W2′ folds into F6—see below—so the sole open premise is F6.)

The keystone, sharpened. Two cold attacks on F6 turned the vague "bounded width" into a precise dichotomy. The naive ways to force width—shell half-graphs, prefix pumps, single-chain PO-gaps—all *collapse* to finite catalogues (they have $O(1)$ separator cover, a carried memory slot suffices). The sharp obstruction is an unbounded family of M_n *identity-selector minors*: pairs (L_i, U_i) each separated by a *private* selector A_i and by no other, so the selector incidence matrix is the $n \times n$ identity—private matching of unbounded hitting number. In plain terms: n separate "this-region-not-that-one" witnesses, no two of which may be merged because each is the *unique* reason some specific pair stays apart—so a finite catalogue, which can only hold boundedly many such witnesses, is overrun exactly when a concept forces n of them for every n. F6 fails iff some concept *forces* such minors; F6 holds iff none does. And W2′ (Definition 11), which looked like a second obstacle, was shown false in its coarse form but repairable by a state refinement whose size is bounded *by F6*—so it *folds into* F6, leaving a single open question (`fable5_width_barrier/`, `cold_review_f6_w2prime/`; probes `wp35`–`wp44`, incl. `wp44`: arbitrary RCC5 glue steps *can* have unbounded separator cover, so any F6 proof must use $\mathcal{ALCI}_{\mathrm{RCC5}}$-forceability, not RCC5-consistency alone).

The problem is Π^0_1, and F6 weakens to its qualitative core. One further observation—new to this project—sharpens what F6 must deliver. As §5 sets up, the semantics is finitely *first-order* axiomatizable (the theory T_{RCC5} plus the standard translation of C_0), over *arbitrary* carriers—exactly what the carrier-polymorphic satisfiability of the Lean development means.

Concretely this is a *computable reduction to plain first-order satisfiability*: each C_0 maps to the single sentence $\bigwedge T_{\mathrm{RCC5}} \wedge \exists x\, \mathrm{ST}(C_0, x)$, which has a first-order model iff C_0 is satisfiable. And first-order satisfiability is itself Π^0_1—the co-r.e. side of Church's undecidability theorem, which is Gödel completeness in action: unsatisfiability equals refutability, and FO refutations enumerate. So satisfiability is *in* Π^0_1—and this is an upper bound only: we establish *membership*, not Π^0_1-hardness (a *decidable* problem is also in Π^0_1). It shares its *ceiling* with the domino problem, but we do not show it as hard, so the "knife's edge" (§1.2) is a suggestive analogy here, not a completeness result. The *reduction* alone would not even fix this ceiling: a non-first-order model condition—well-foundedness, finiteness, anything genuinely second-order—could push it higher; the finite axiomatization is what keeps the reduction a single sentence.

The consequence for the keystone is best seen through the missing *dual*. A Π^0_1 problem is decidable iff it is *also* recursively enumerable, and completeness hands us one such side—enumerability of UNSAT—for free. The classical way to get the *other* side, enumerability of SAT, is the *finite model property* (enumerate finite models), the textbook route "recursively axiomatizable + finite model property ⇒ decidable"—a principle report7 itself already invoked, en route to a bounded variant of these logics [8]. But $\mathcal{ALCI}_{\mathrm{RCC5}}$ has *no* finite model property (§5: the ascending PP-tower needs an infinite model), so that route is shut, and enumerating SAT the honest way means enumerating *finite certificates for possibly-infinite models*—which is precisely *qualitative* F6: every satisfiable concept has *some* finite certificate, *with no computable bound demanded*. One dovetails such a certificate enumeration (its checker the already-certified `finAccept_sound`) against the enumeration of FO refutations; on every input, one side must eventually fire. So F6 weakens from its *quantitative* form (a live-width bound *computable* from C_0—the target behind most of the width-accounting defects the reviews found) to the *qualitative* core just isolated, and the computable bound then returns for free *a posteriori*.

The dovetailing reduction is itself machine-checked (`formal/SemiDecidability.lean`, zero `sorry`), and the FO transcription it rests on—that the axioms above carve out exactly the Lean frame conditions, and that the standard translation matches the certified satisfaction relation—is cross-checked in `wp84` (exhaustively for domains up to three points). Two honest caveats: the qualitative core of F6 is *untouched*, so this reshapes the target rather than hitting it; and the Π^0_1 placement adds one standard external theorem (Gödel completeness) to those the argument already invokes (the RCC5 patchwork property; parity-automaton emptiness in the historical automata route). No complexity bound is claimed or implied.

7.6 The regular-cover pivot

Idea. A reframing suggested by the human author: a finite *quotient* of a PP-chain is invalid (a PP-cycle contradicts EQ-identity), but a finite *generator* is not—a state with a PP-self-loop *unfolds* to an infinite chain of distinct copies. So replace finite quotients by *regular covers*: finite presentations whose infinite *unfoldings* are the models, in the spirit of the two-tier quotient but for the full logic. This finer invariant can represent exactly the patterns that fool a width bound—identity selectors ($i = j$ via address equality), half-graphs ($i < j$ via an order test)—finitely.

What it produced. A complete *local* algebra of strong-EQ RCC5 networks, one piece of which is machine-*certified*:

Theorem 13 (RCC5 normal form)**.** *A strong-*EQ *atomic RCC5 network is composition-closed if and only if it is an* ordered-disjoint structure*:* PP *is a strict partial order,* DR *is a symmetric, irreflexive, downward-closed disjointness (if $x\,\mathsf{DR}\,y$ and $x' \leq x$, $y' \leq y$ then $x'\,\mathsf{DR}\,y'$), comparable pairs are never disjoint, and* PO *is the residual. The forward direction (every RCC5 network is ordered-disjoint) is* machine-certified for arbitrary domains *(`formal/RCC5NormalForm.lean`, axioms `propext` only); the converse is exhaustively machine-checked up to size four (`wp47`).*

With it come free amalgamation of ordered-disjoint structures, an exact one-point extension criterion, and exact uniform/non-uniform cross-policy characterizations—together *exhausting* the local algebra (`rcc5_local_amalgamation_theory`, `regular_cover_route_pivot/`, probes `wp45`–`wp83`). The route independently re-derives Theorem 10.

Why it is a better map, not a shorter path. Its own keystone (a bounded coherent regular-cover property) is F6 relabelled, and the route concedes, after exhausting the local algebra, that the remainder "cannot be a purely local RCC5 composition problem." Because RCC5 relations are total, every model's Gaifman graph is complete, so a one-dimensional PP-chain already has unbounded clique number and treewidth; no off-the-shelf guarded or bounded-treewidth cover theorem applies, and the keystone would need a bespoke cover theorem for complete edge-coloured regular structures. The pivot is the right vocabulary for a future attack—but it does not move the summit.

8 Current implementation: reasoners

The project ships working reasoners, on the same foundation as the proof attempts but validated empirically rather than proved.

The cover-tree tableau. `src/cover_tree_tableau.py` (≈ 350 lines) is an operational concept-satisfiability procedure with exact-occurrence blocking, resting on the split-forest normal form and patchwork completion (§7.3). It is cross-validated against an independent quasimodel oracle on 911 concepts (zero mismatches after the known blind spot is excluded) and against an independent model verifier (`src/model_verifier.py`). It is trustworthy on satisfiable inputs; its one known limitation is the strong-EQ PO-loop, decided correctly by the tableau but not by the quasimodel oracle. It is strong evidence, not a proof: no completeness theorem is claimed for the implementation.

Evidence statistics, and the absence of a counterexample. The empirical case is substantial, and it is worth stating precisely because it is the strongest thing short of a proof. The two reasoners agree on a 911-concept cross-validation suite (zero mismatches after the single known PO-loop blind spot of the quasimodel oracle is excluded), and the cover-tree decomposition passes 775/775 of its structured tests. We state its scope honestly: the two implementations are genuinely different algorithms (tableau versus type elimination) but share the composition table and test harness, so this is cross-validation with some common-mode risk, not two wholly independent oracles; and the model verifier confirms explicit *finite* SAT witnesses, not UNSAT verdicts or infinite-model SAT. Beyond the reasoners, some eighty self-contained verification work packages (`verification/python/`, WP1–WP87) cross-check the finite combinatorial content of every proof route against the RCC5 composition table *re-derived from finite set semantics*, so the underlying algebra cannot be fudged. Representative results: the RCC5 patchwork property holds on 63,219 two-bag amalgamations plus 300 tree-shaped patchings with zero failures (WP14); the finite acceptance mechanism matches the cover-tree tableau on ten diagnostic concepts plus a 148-concept random sweep with zero disagreements (WP16); the cluster-quasimodel decision layer matches the tableau on 11/11 diagnostics plus a 200-concept sweep (WP26); the RCC5 normal form agrees with brute-force composition closure on all networks up to size four (WP47); and the Lean development builds with zero `sorry` on every run (WP34). Most tellingly, *across all thirty rounds and sixteen adversarial reviews, no accepted-but-unsatisfiable concept was ever found*: every defect the reviews turned up was a gap in an *argument*, never a semantic counterexample to the intended theorem (the absence of a hybrid $\downarrow$-binder—a way to name the current point and reuse that name inside nested modalities—appears to block concept-level exploitation of the seams the reviews located; note that *nominals* themselves *are* definable under strong EQ, so it is the binder, not nominals, that is missing). This is not a proof—no amount of it can be—but it is the reason we describe the decidability argument as *strongly supported.*

A GIS example spanning 23 years. The tableau recomputes the GIS concept taxonomy of Wessel's original report [8] (its Figure 6, the hand-drawn 2003 original, reproduced here as Figure 4): all 21/21 subsumptions are recovered (`src/gis_taxonomy.py`), and the reasoner emits its own rendering of the resulting hierarchy, shown as Figure 5. A taxonomy computed by hand in 2003 is reproduced, in 2026, by an AI-written reasoner resting on the split-forest foundation—and the two diagrams agree edge for edge.

9 On agentic AI usage

We report candidly on the workflow, because the methodology may outlast the mathematics.

The loop. The human posed the problem, supplied the central intuitions and directions (the split-forest decomposition, the cluster/loop and regular-cover reframings, the honesty discipline),

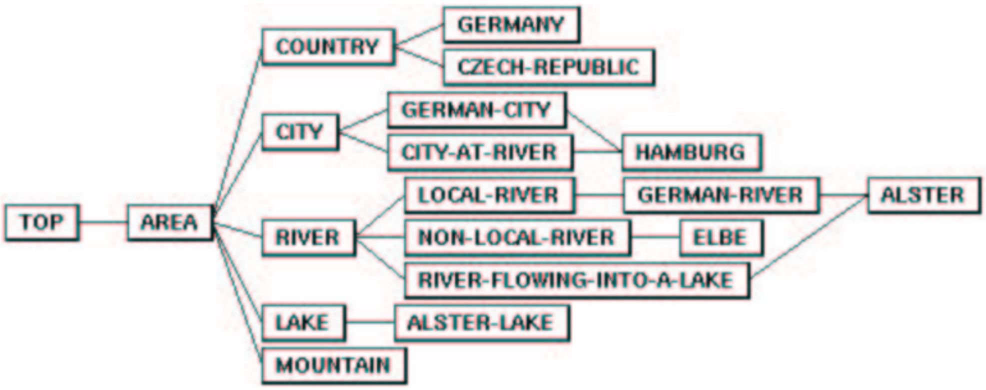

Figure 4: The GIS concept taxonomy as drawn by hand in Wessel's 2002/2003 report (report7, Figure 6): the original.

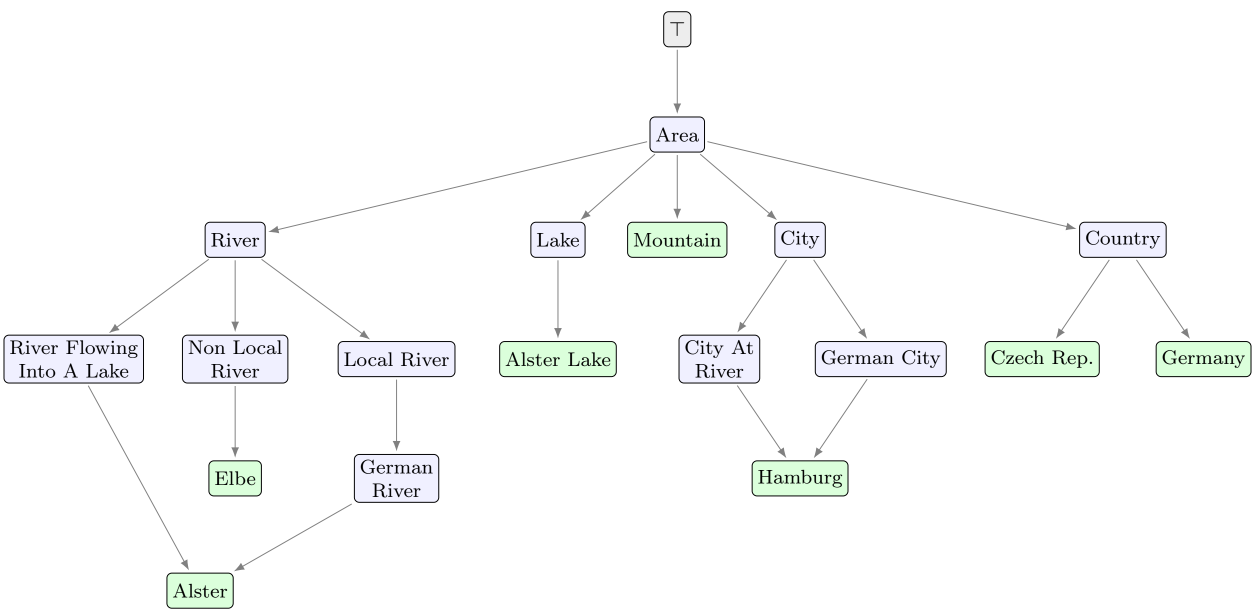

Figure 5: The *same* GIS taxonomy recomputed in 2026 by the cover-tree tableau reasoner (`src/gis_taxonomy.py`), rendered directly from the reasoner's own output (arrows point from the more general concept to the more specific; green = individuals). All 21/21 subsumptions match Figure 4 edge for edge. Note that *Hamburg* (a German city at a river) and *Alster* (a German river flowing into a lake) each have two parents: the taxonomy is a directed acyclic graph, not a tree—exactly the multiple-inheritance the reasoner must get right.

and after each round reviewed, corrected, and re-prompted. An AI proposed a proof or a repair; a *fresh* AI session—with no memory of how the proof was built—then reviewed it adversarially. Repeat.

Division of labour, and a calibrated comparison. The roles were asymmetric: one model (GPT-5.5) chiefly authored proofs, another (Claude) chiefly implemented, verified, audited, and documented, and the human supplied the direction and the original insight. It is tempting to read the review record—each model repeatedly finding defects in the other's drafts—as ranking them, and we resist that inference: it is confounded by role (proof-authoring and cold-reviewing were never swapped), by model version (both lineages advanced mid-project), and by task (cold-reviewing a finished manuscript is a different, arguably easier job than authoring one). The robust finding, independent of which model is sharper, is that *neither is self-sufficient*: both write fluent long-context formal mathematics, both fail to reliably self-review, both at times declared results "closed" prematurely, and neither produced the decisive original idea. What carried the project was the loop, not any single model.

An auditable record, for honest attribution. Because "who contributed what" is exactly the question that AI-assisted work makes fraught, we kept the entire, timestamped conversation log public [60], alongside the round-by-round manuscripts and a superseded-work archive (`OUTDATED.md`). A reader can trace each idea, defect, and repair to its origin—human or model—rather than take our word for the division of labour above. We treat this as part of the methodology, not an afterthought: a human director should not quietly absorb credit for a model's mathematics, nor should a model's fluent prose obscure the human's insight, and the only reliable guard is a record open to inspection.

Cold review as a standard step. The single most transferable finding: a *cold* reviewer consistently found defects that a *warm* reviewer (primed by the construction) missed—including, on this project, unsoundness proofs and scope errors that two hot audits had passed over. Across sixteen reviews, a defect or overclaim was found in all but two. We recommend cold, cross-lineage adversarial review as a default intermediate step in AI-assisted mathematics, not an optional one.

Don't trust, verify. Every load-bearing claim was cross-checked against something mechanical: self-contained Python probes that re-derive the RCC5 composition table from finite set semantics (so the algebra cannot be fudged), and ultimately the Lean kernel. Overclaiming was treated as the cardinal sin—and it happened: the author model twice overstated what the Lean established, and a cold review caught both; the claims were withdrawn. A sixteenth (cold) review of this very manuscript then found a *definite* error—an illustrative one-point-extension example whose claimed contradiction does not hold, the two triangles in fact forcing the single relation PO by *joint* constraint (Remark 2)—and, with it, that the informal "PO is never forced" intuition had been stated too strongly. Both were corrected, and the distinction between single-step and joint forcing is now made explicit throughout. That an error of this kind survived into a tagged manuscript is exactly why the standing label is *not certified*: the machine checks, not the model's (or the author's) confidence, are the ground truth.

What frontier models did well, and did not. Both models sustained long-context, rigorous, multi-round formal development and produced genuinely useful adversarial reviews and machine-checkable artifacts. What was *not* demonstrated is the decisive novel idea—no model cracked F6, and the honest expectation is that none will by the incremental means used here. The models excel at rigorous local formalization and at finding flaws; the genuinely new keystone is where human insight (or a qualitatively different search) is still required. We regard the AI contributions as co-authorship-grade in a human collaborator; we also regard the human's role—problem, intuition, direction, and the discipline of honest labelling—as indispensable.

A worked example: a seed became a fragment. The clearest single piece of evidence for the value of the loop is not the F6 work but the ∀PO-free fragment (§7.2), and it is worth being precise about the *shape* of that contribution. The human author supplied a seed whose consequences he could not compute in advance: collapse same-type occurrences on a PP-chain into a kernel node with a reflexive PP-loop. The models then did the part that is tedious and error-prone by hand—the exhaustive algebra (reflexive PP is self-absorbing; PP-cycles are impossible; single-type chains collapse but multi-type ones do not), the two-tier construction that routes around the obstruction, and, the decisive step, locating exactly where the idea *breaks*. The PO gap was in neither the seed nor the first construction; it surfaced under cold review, and the result crystallized precisely at that boundary—the fragment is defined by exactly where the seed stops working. No single participant produced this: human intuition chose *which* idea to chase, the construction turned it into a real object, and the cold review caught the gap the construction had missed. That is the loop of this paper exhibited on a concrete result rather than asserted—and it is the honest, bounded form of the title question. Not that agentic AI settled the open problem, but that a human–model loop turned a vague structural hunch into a decidable fragment with a machine-checked soundness core that *none* of the participants, human or model, would have produced alone. One caveat we keep in view: this is a new result *of the collaboration*; its novelty relative to the prior description-logic literature we have not vetted, and a priority claim would need that check first. The *process* contribution, however, is unambiguous.

10 Summary and conclusion

What we learned. A twenty-five-year-old open problem has been *reduced*, honestly and largely machine-checkably, to a single well-posed lemma (F6, bounded live width), which is provably the same fact that blocks a proof of undecidability. The local algebra of RCC5 is exhausted and, in one case, certified. Adversarial cold review plus machine verification is a repeatable methodology for this kind of work.

What would move the needle. F6 is a *definability* question—can a formula *force* unbounded rigid horizontal structure?—so the tools that fit are model-theoretic: locality (Gaifman/Hanf), Ehrenfeucht–Fraïssé / back-and-forth over bounded tuple profiles, the composition method, and bounded-model machinery. The concrete positive target is a *selector-sharing lemma*: that many same-profile selector gadgets must coalesce, so unbounded identity-selector minors cannot be forced. The concrete negative target is a concept forcing such minors (which would prove undecidability). Ordinary locality alone is insufficient because horizontal degree is unbounded; this is the "correct room," not a solved problem.

The chances of closing F6. We are frank: they are low, and re-treading the ground covered here will not change that. F6 is the crux on which this architecture turns—though, strictly, only F6 $\Rightarrow$ decidability is established, and a *failure* of F6 would refute *this* certificate route without by itself proving undecidability (the logic could still be decidable by some representation this width measure does not control). The two independent cold attacks converged on the same wall—evidence that the problem is correctly *mapped*, not that it is about to yield. It would be a poor use of effort to reconstruct the split-forest, automata, Lean, or regular-cover developments; the value of what remains is a genuinely new definability-theoretic idea, or a machine-assisted search of a different kind (below).

The wins. Independent of F6, this project produced: (i) an unconditionally *decidable fragment*, the $\forall\mathsf{PO}$-free concepts; (ii) a machine-*certified* RCC5 normal form; (iii) a machine-checked *conditional decidability theorem* (F6 $\Rightarrow$ decidable, soundness certified, resting on the F6 oracle); (iv) a *working, empirically validated decision procedure*—the cover-tree tableau (§8), cross-validated against an independent quasimodel oracle on 911 concepts with *zero* mismatches, reproducing 23 years later the full GIS subsumption taxonomy of Wessel's original report (21/21), and across the entire campaign never once producing a decision contradicted by any other oracle or by a hand analysis—*no counterexample was ever found*; (v) a sharp *map* of the problem with a named toolbox; and (vi) a candid *methodology* for adversarial AI-assisted mathematics.

Two of these are the project's flagship artifacts, and they are complementary rather than ranked. The Lean development (items ii–iii) is a *certified* surround: the kernel guarantees the soundness pipeline and the abstraction's faithfulness contain no gap. The reasoner (item iv) is the *empirical* counterpart: a real algorithm that decides concepts, has been run on thousands of them, agrees with an independent implementation everywhere it has been tested, and reproduces a two-decade-old published taxonomy. Neither proves decidability—but that a running procedure and a machine-checked proof both point the same way, and that no adversarial search over the whole campaign ever exhibited an accepted-but-unsatisfiable (or rejected-but-satisfiable) concept, is the strongest *evidence* short of a certificate that the conjecture is true.

Is this a scientific contribution? We believe so, with the labelling kept honest: not a resolution of the open problem, but its reduction—checkably, and with a certified surround—to an irreducible core, plus one unconditional fragment, one certified lemma, working tools, and an honest account of an AI-assisted attack and its dead ends. That is an accurate and, we think, respectable terminal state for a discussion piece on a problem open for over two decades.

If nothing else: a demonstration, addressed to our peers. Whatever the eventual fate of the mathematics, this report is at minimum an expressive demonstration of what frontier AI models could do in serious formal research at the time of writing (July 2026): sustain a months-long, adversarially-reviewed attack on a twenty-year-old open problem, author and repair proofs across some thirty rounds, catch one another's defects with machine-checked witnesses, drive the surviving core into a kernel-checked Lean development, and ship working, validated reasoners along the way. Every description-logic researcher—indeed every logician and mathematician doing serious work—should be aware that this is what the instruments can now do. Ignoring these capabilities does not advance science, and declining to use them is, we think, simply unwise. Informing our peers of exactly this was itself a major purpose of the present report—a message first offered through the community's own venue and, as Appendix A documents, rejected on grounds that engaged neither the mathematics nor the message. We record it here again, in the open.

Outlook: the human is the bottleneck. Although the key insights, intuitions, and directional choices came from the human, each round still required the human to orchestrate, review, and rewrite the prompts—and at that cadence roughly eighty rounds consumed about four months. Autonomous sub-agents *directing* the research could plausibly automate the diminishing-delta local work (formalization, local lemmas, probe-driven counterexample search) and remove that bottleneck. The cost may be prohibitive and it sacrifices direct oversight, but it is worth attempting for one reason this project is unusually well-suited to it: the objective is clearly defined and *machine-verifiable in Lean.* A closed, adversarial loop—propose, formalize, let the kernel accept or reject—has a built-in ground-truth oracle, exactly the guard the manual cold-review methodology provided. The honest caveat: Lean verifies *proofs*, not the *truth of a conjecture*, so such a loop can tighten the certified surround but the summit (F6) still needs a genuine idea; orchestration alone is unlikely to supply it.

Materials. The machine-checked development is `formal/Round19Transport.lean`, `formal/RCC5NormalForm.lean`, `formal/ForcingReduction.lean`, `formal/SemiDecidability.lean`, and `formal/POFreeLift.lean` (see `LEAN.md`); the status report and the F6/W2′ analysis are in `papers/fable5_width_barrier/`; the decidable fragment in `papers/two_tier_quotient_ALCIRCC5` and `papers/po_free_fragment_ALCIRCC5`; the certified local theory in `papers/rcc5_local_amalgamation_theory`; the reasoners under `src/`; the probes under `verification/python/`; and the full audit trail in `CONVERSATION.md` and `OUTDATED.md`. The repository is at https://github.com/lambdamikel/alcircc5.

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A On the DL 2026 rejection

An extended abstract of this work was submitted to the 2026 International Workshop on Description Logics (DL 2026) and rejected. We record the episode—and dissent from the decision—not merely out of grievance but because it bears directly on this paper's subject: how a research community engages with an AI-assisted attack on one of its own long-standing open problems. The submission was labelled exactly as this paper is (*strongly supported, not certified*, an AI-assisted discussion piece), and its stated motivation, twice, was to raise awareness of the emerging capabilities of agentic AI for description-logic research. We contend it was rejected on the wrong basis.

1. **The technical objections targeted the optional appendix.** DL 2026's own guidelines state that the appendix "will not be included in the proceedings" and is reviewed only at reviewer discretion. Yet nearly every concrete objection—missing definitions, a broken cross-reference, an under-specified lemma—concerned appendix material, not the extended abstract that is the reviewed artifact.

2. **The reviews attacked a position the paper did not take.** The abstract called the proof a "candidate," wrote "if found correct by human experts," and carried an explicit non-verification disclaimer. It was nonetheless assessed as a settled-theorem submission and rejected for not clearing that bar—a strawman.

3. **Honest hedging was quoted back as a fault.** The very disclaimers included to avoid over-claiming were cited as grounds for rejection. Punishing calibrated uncertainty inverts the incentive peer review should create.

4. **No review engaged the actual thesis** (agentic AI's capabilities), nor the concrete contributions the submission *did* present: a working cover-tree tableau reasoner reproducing the 2003 GIS taxonomy, and an unconditionally decidable fragment. (The split-forest normal form, the reduction to a single keystone, and the machine-checked Lean soundness reported in the present paper came later, in the months *after* the rejection—so they cannot have been the basis for it.)

5. **The verdict was a value judgment, not a technical assessment.** The three reviews largely agreed on the defect inventory but split on the verdict—one reviewer believed the result true and named the right approach, yet recommended reject on presentation; another recommended accept. The decision was escalated to the Steering Committee, and the covering letter offered that the paper "could be a good scientific contribution if it … focus[ed] on the advantages and disadvantages of using LLMs … rather than claiming decidability results." But that discussion—the advantages and disadvantages of agentic AI for this kind of research—is exactly what the submission's main text *did* provide, alongside the concrete results above; it simply was not engaged with. The "acceptable" paper being described was, in substance, the paper already submitted. The objection was to the framing and the precedent, not the substance.

6. **A disclosure paradox.** The integrity choice—front-loading the AI co-authorship, in the title and from page one—was precisely what made the submission unacceptable, while the offered path to acceptance was to write it in standard single-voice style with a tail-of-paper note. A process that rewards understating AI's contribution and penalizes overstating it selects for under-disclosure, not for science.

We grant one point: a single introductory sentence ("We hereby claim: $\mathcal{ALCI}_{\mathrm{RCC5}}$ is decidable") read more assertively than the surrounding, carefully hedged text supported, and a constructive "tone this down and resubmit" would have been met immediately. That does not carry the verdict. There is, finally, a recursive irony: the rejected abstract had warned, by name, about the failure modes—network effects, not-invented-here bias, dismissive reviewing—that the rejection then illustrated. The deeper concern is not one abstract's fate but that agentic AI now lets ideas reach decades-old open problems from outside the established networks, and a review process that answers this by rejecting on framing while ignoring substance constrains rather than protects science. Peer review exists to surface and stress-test new ideas, not to keep them out; when it functions as the latter, it has stopped doing the job the community delegated to it. The full extended abstract, the three reviews, and a detailed analysis are public (`dl2026-rejected/`; the open letter is `open_letter.md`); readers are encouraged to form their own view.